Observers Who Read Their Universe into Existence: Finite Consensus and the Reconstruction of Standard Physics

Bernhard Mueller[1,2,*] Mario Poneder Alexander Osika[3] Kai Xue
Ben Cassie Peter Nguyen Jinwook Kim[4] David Matscheko
Daniel Ari Friedman[5,6] Jonathan Hill William T. Glynn Maarten Antonie Visser
Kale Arnav Anirudha Brieuc de La Fournière

November 7, 2026 · Paper release: r2062

[1]Pragma Research Inc. [2]Information Physics Institute [3]EtherWorks [4]Oraclizer Labs [5]Active Inference Institute [6]FractiAI
[*]Corresponding author: bernhard@floatingpragma.ai

Abstract

Why is there anything at all? An equation can describe a real structure, but consistency alone does not establish that it exists. Observer Patch Holography proposes a closed, self-referential history whose inhabitants recover the rules producing their records and realize those rules within that same history. Consistency has a local mechanism: finite observers compare overlapping records and repair disagreements. A fact is public when its record survives every authorized comparison and repair. We prove a canonical finite consensus theorem. On a twelve-port carrier, repair has exactly three slow directions; a faithful complete current implementing the carrier's rotations internally has the Standard Model gauge Lie type. A supplied complex algebra of states and effects yields quantum measurement identities, and fair-coin repair gives the entropy laws on binary populations. A specified finite source connects protected defect records, currents, certified charged spectra and interacting energy levels. Controlled free-field constructions and conditional geometric response laws connect this finite framework to field theory and the Einstein equation. The central proposal is that existence, objectivity, space, matter and gravity belong to one self-consistent process. Its identification with our universe requires a common interacting chiral $3+1$-dimensional limit with the observed particle spectrum, responsive geometry and independent empirical agreement.

Keywords: foundations of physics; Observer Patch Holography; finite consensus; self-reference; quantum records; gauge structure; particle physics; Einstein dynamics.

1 Introduction

Mathematics describes structures that may be physically real. An equation for a wave or a fixed point states relations; a physical realization gives those relations observable content. The existence question asks whether the whole process could admit an explanation of this kind. A brute initial fact ends explanation at an unexplained starting point. A chain of prior causes moves the same question to an earlier link. Neither option is a logical contradiction, but neither explains the whole on its own. OPH asks whether the complete history can close on itself instead.

The inversion is that a description of the world is itself an event in the world. Observers infer rules from public records, construct a process that implements those rules, and belong to the history that the process describes. The self-reference has content only if one specified history actually satisfies this return condition and reproduces the observations made within it. This is the governing question of the paper. Consistency constrains the construction; it does not make an arbitrary equation into an existent universe.

1.1 Mission

The mission of OPH is a consistent and complete, formal and rigorous theory of reality. Its proposed complete history has no external starting point: it closes on itself. A successful theory must meet four conditions.

1. **Consistent.** The loop closes. Its fixed point exists, is constructed explicitly and is machine-checked, and a theory with a model contains no contradiction.

2. **Explicit.** The complete recovered specification fixes its consequences. Every design choice and calibration is declared; quantities claimed as independent predictions are not inserted through those inputs.

3. **Complete**, in the sense Einstein, Podolsky and Rosen gave the word [23]. Every element of physical reality has a counterpart in the theory, and every measured quantity is derived.

4. **Correct.** Derived values agree with measurement, and new predictions are frozen before they are tested.

The mission ranges from the existence of an inhabited history and the possibility of experience to the number of spatial directions, quantum statistics, forces, particles, gravity and cosmology. This range specifies what the proposed account must explain; it does not turn a finite model or a conditional theorem into a derivation of every item. In particular, a three-dimensional carrier response needs a physical position map, and an operational observer needs an additional account before any claim about phenomenal qualities follows. Table 2 gives these distinctions alongside the physical results.

OPH does not claim to be the only theory with these properties. The four conditions organize this paper. Consistency is tested through the closure hypothesis and the executable model (Section 2) and through the machine-checked results of Section 14. Specification and parameter ancestry are tested through the closure coordinates of Section 11.3. Completeness is the ledger of realized results and physical obligations in Section 12. Correctness is decided by the frozen predictions and tests of Section 13.

1.2 Consistency through local repair

Local repair is the engine of the proposal. A finite observer has no global view. It holds a patch of local state, reads the boundary shared with neighbors, retains records and repairs disagreements while protecting what has been recorded. A fact becomes public when every authorized comparison preserves it. The finite consensus theorem gives such records a canonical normal form independent of the repair schedule. Observer Patch Holography (OPH) asks whether the world we measure can be reconstructed from this kind of compatible local access. Quantum probability, Lorentzian kinematics, gravitational dynamics and gauge structure enter standard physics through distinct mathematical inputs [24, 25, 26, 60, 67]; OPH tests whether one observer architecture can carry

their public readouts. Space, time, matter and classical objects are effective descriptions of what observers can measure and jointly interpret. In this interpretation, spacetime geometries and quantum field theories describe relations among compatible records; the proposal assumes no prior observer-independent continuum.

The twelve-port carrier makes this idea concrete. Optimal readout of a three-dimensional response selects its incidence, and its normalized repair response has a three-dimensional slow quotient, a candidate carrier for spatial position. Under the further premise that a faithful complete reversible current implements the carrier's rechartings internally, its Lie algebra has type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$. Three directions and the Standard Model gauge type thus arise as different constraints on one boundary network's response. Identifying those structures with physical position and laboratory currents is a further test.

At continuum scale, consistency links matter and geometry. If one history supplies conserved stress and makes every local null direction obey the same curvature–stress balance, tensor completion yields the Einstein equation up to a cosmological constant. Obtaining that balance, the entropy response and a common physical geometry from the finite source is a separate requirement. OPH's unifying hypothesis is that the full dynamics of nature are the conditions under which one self-reading process keeps its records coherent through local repair.

The intended argument runs from specified repair operations to compatible joint records, and from those records to a common operational description of space, time and matter. A position is what the spatial readouts locate; a duration is what calibrated clocks compare; a particle is a persistent excitation with the observed transport, charges and interactions. The target is one such description with three spatial directions and local time, including its quantum fields and gravitational response. The finite constructions below establish parts of this chain; the common physical $3 + 1$ limit is the remaining conclusion to be demonstrated, not an input hidden in the word "observer."

OPH makes one proposal about the whole, closure: one complete history is both a timeless self-referential structure and its own internal computation, and observers inside it recover its specification and realize it (Section 2). The executable-model hypothesis is the concrete and testable form of that proposal: one specified local repair and memory process must supply records, clocks, fields, matter, energy and responsive geometry together. This paper presents OPH as that single theory, organized by its dependencies: each layer states a hypothesis, proves finite theorems on displayed premises, and passes a conclusion upward only when the executable model realizes the premise on the same process.

1.3 Scope and order of the argument

The program has four logically different levels. Closure proposes that an inhabited history contains a recoverable account of its own operations. A source specification states the carrier, observer access, repair, record and state laws. Finite theorems and executable certificates determine what that specification does. Physical realization asks whether one such source has the observed fields, particles, clocks and responsive geometry. The implications run in that order; Table 1 states the result at each level.

The finite results make OPH a concrete reconstruction program. Their composition into the observed universe is a separate mathematical and empirical claim. The specialist papers give the longer proofs, source executions and failed attachments [12, 11, 9, 68].

Finite theorems, complete source executions and physical interpretations answer different questions. We state each result's domain and supplied inputs where it is used; the detailed claim classes and verification methods are collected in Section 14.

Table 1: The OPH program and the status of its main implications.

Level	Result on its stated domain	Physical requirement
Closure	Finite consensus has an exact fixed-point packet theorem; a universe-level self-recovering history is proposed	A specified history and realization map satisfying the closure condition
Records and carrier	Canonical public records, conditional quantum identities, finite entropy laws of repair and an exact rank-three slow response	A common physical position, clock, energy and preparation map
Gauge and matter	Complete-response premises force the Standard Model Lie type; specified finite operators join currents, modes and mass coefficients	An interacting chiral field with observed charges, stable species and pole masses
Gravity and cosmology	Null tomography and entropy premises imply the Einstein equation; finite capacity and pixel equations give conditional relations	The same source must supply conserved stress, responsive geometry, vacuum, scales and prospective observational matches

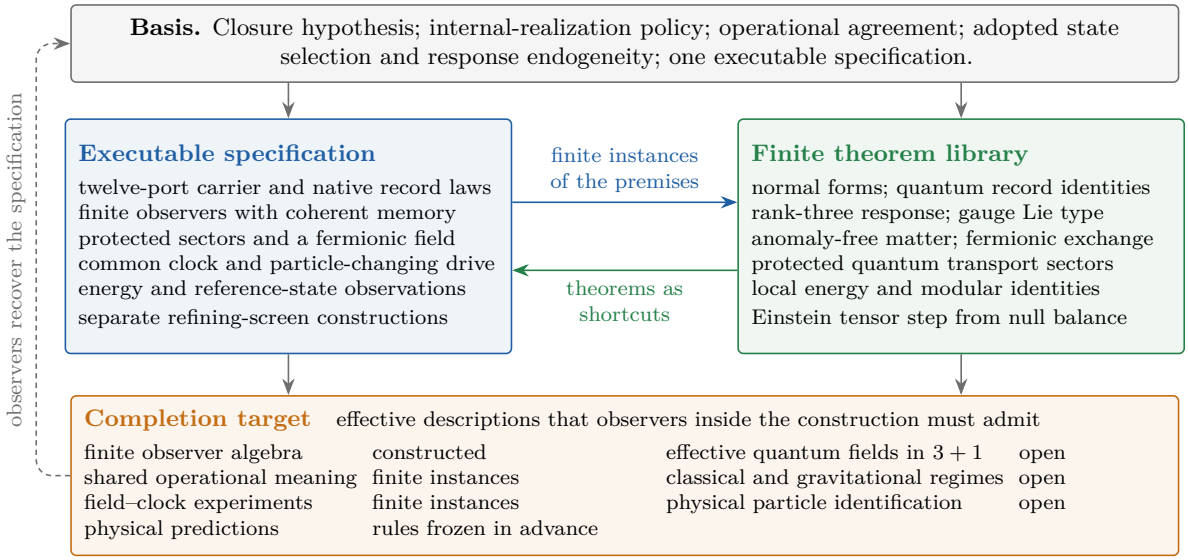


Figure 1: The proposed chain from protected records to physical laws. The finite theorems and source executions constrain one specification; physical conclusions require compatible realization maps. The dashed arrow is the closure proposal: observers recover the specification from their records.

"Source-executed" means that an operation is compiled and run using the declared native preparation, repair, memory and readout services, with its complete outputs and resources accounted. "Source-only" concerns the provenance of its parameters: no empirical physical input occurs anywhere in their dependency chain. These are different properties. A source-executed field may use empirically calibrated Yukawas, mixing parameters or clock units; reproducing those inputs verifies the implementation, not an independent prediction.

Definition 1.1 (Physical realization map)**.** A physical realization map assigns finite public records,

event data, currents, modular generators, and scales to physical records, spacetime events, laboratory currents, clocks, and units, preserving the algebraic operations, overlap restrictions, causal order, normalizations, and refinement relations consumed by the conclusion in question. A resemblance of spectra, dimensions, or symmetry names is insufficient.

The paper gives one mathematical representative of each step: the native repair rule, the consensus normalizer, the spatial and gauge reductions, the finite matter attachment and the Einstein tensor implication. Its status tables and prospective tests specify the physical conclusions. The companion construction and component papers carry implementation contracts and longer proofs [68, 1, 4, 5, 6, 8, 11, 9, 7, 13, 14, 22, 2].

2 Closure and the source-specification hypothesis

This section states the interpretation and acceptance criterion: closure, the internal-realization policy, the additional state-selection and response-endogeneity principles, and the agreement condition, then explains why closure is tested through an executable model.

2.1 The closure hypothesis

The universe is modeled as a timeless, closed, self-referential structure in which the description and the described system are one object. Let $\mathfrak{U}$ be a complete inhabited history, let $D(\mathfrak{U})$ be an account of its architecture recoverable through completed records within it, and let R be an admissible realization of that account constructed inside $\mathfrak{U}$. The closure condition is

$$R(D(\mathfrak{U})) \;\cong\; \mathfrak{U}, \tag{1}$$

where $\cong$ is agreement in the observable and interventional structure: the records used to recover the description, and the responses of the realization to every admitted intervention, fit the world from which the account was recovered. Written as a fixed point of a universe-level closure operator, the condition reads $T(\mathfrak{U}) = \mathfrak{U}$. Observers infer the specification of their world, build hardware within it, and run that hardware; its execution realizes the same history, including its builders and their hardware. The identification of the construction-side and inhabited-side readings is the hypothesis. In this sense the structure reads itself into existence.

Within this hypothesis, no external computer runs the history and no additional time exists in which the whole is played. Host runtime used to check a model is part of our method of investigation, not an extra time coordinate of the universe described by that model. Repair order is a partial order on commits, an observer history is a chain of completed records, and that order is invariant under strictly increasing regradings, so affine time and proper time require their own calibration. Fixedness of the complete history permits changing states within it: the equation concerns the whole history and identifies no last instantaneous state with the universe.

Self-reference earns content only when the return condition restricts the possibilities. On the real line $x \mapsto x + 1$ has no fixed point, $x \mapsto x$ fixes every point and $x \mapsto (x + 2)/2$ fixes exactly one. A candidate therefore has to specify the history domain and the action of T; existence, consistency and the identification of our universe with a fixed point are questions the equation leaves open. Self-reference alone guarantees neither consistency nor mathematical completeness. Uniqueness is an additional question when it is claimed; it is not a prerequisite for a specified construction to work and match observations. Two finite results give the condition exact content on declared branches. The packet-quotient closure theorem proves that on the declared finite consensus branch the closure map is an affine idempotent whose fixed points are exactly the packets supported on

consensus normal forms [6]. The universe-level record-closure equation $N = \log M_0(\mathfrak{U}_N)$, with $\mathfrak{U}_N$ the regulated observer system at capacity budget N and M_0 the correctable-code capacity of its terminal public record, is posed exactly, and the all-rung arithmetic proves that the bounded completion class defined by base agreement, positivity and the carrier bound does not identify it [8].

2.2 No outside

A self-contained realization must account internally for its operations and information access. This motivates a justification policy and two further constitutive principles; it does not logically derive those principles.

Justification policy. A premise is justified operationally when observers recover it and the specification implementing it runs and reproduces their measurements. The carrier, observer algebra, repair law and closure coordinates must meet that test. Their specification can be reconstructed from records or proposed as a design. In particular, using a twelve-port carrier does not require a theorem that every possible realization must have twelve ports. A construction choice needs no uniqueness theorem unless uniqueness is claimed. Closure neither exempts a candidate from empirical tests nor proves its antecedents jointly realizable.

State-selection principle. On the declared feasible family, reference, observer cover and positive weights, adopt the information projection minimizing $\sum_p w_p D(\rho_p \| \tau_p)$. This is premise A3, motivated by least-informative inference but separately adopted when used as a physical selection law. The closure equation supplies no entropy functional or reference; even its identity-map specialization fixes both uniform and nonuniform states. Under A3, every minimizing local density has the union of feasible supports, and a feasible failure witness $b > 0$ gives an explicit selected-probability floor [9].

Response-endogeneity principle. The complete-response branch additionally requires each proper carrier recharting to be implemented by the connected response group, up to a unitary centralizer acting trivially on the response algebra. This is the endogeneity clause of A2, not a consequence of internal implementation alone. For example, twelve diagonal phase generators on $\mathbb{C}^{12}$ admit internal port permutations in $\mathrm{U}(1)^{12} \rtimes A_5$. These permutations act nontrivially on the abelian current algebra, whereas its inner automorphisms are trivial. This closed operation system fails the stronger endogeneity premise; it shows why that premise must be supplied. Theorem 8.1 uses it to select $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ under the stated complete-response hypotheses.

2.3 Agreement and the meaning of "our"

The justification policy fixes the operational test. Agreement says what counts as shared. Observers assign the same operational meaning to accepted data on shared boundaries, naturally under every restriction, recharting and refinement, and a record is public when it survives every authorized comparison (Theorem 4.4). "Our universe" is the world that this public record describes. Agreement in this sense is a definition and an acceptance criterion: it requires interpretation maps fixed independently of validation outcomes and preserving the declared joint and conditional statistics, including quantum correlations. It does not require a noncontextual table of outcomes for incompatible unperformed measurements. Agreement here is compatibility under specified comparisons and interventions, not a vote among observers or assent to a convenient interpretation. Its content

includes the joint probabilities and the response to changed preparations. Calling space, time and matter operational descriptions does not make these constraints optional: once established by compatible records, they are public physical facts. The equations describe those relations and do not by themselves assert an additional material substrate beyond the specified process. The theorem library calls the interpretation clause of this condition A2. The basis reference distinguishes the closure hypothesis, agreement, adopted selection/endogeneity principles and executable specification, with the library's premise statements and their dictionary [99].

2.4 An entry point without a starting point

The exposition of this paper enters the loop at the executable specification. That choice reflects where observers stand, holding records and inferring, and it makes the account concrete. Closure states the internal realization contract; the adopted principles and specification then carry their own mathematical and empirical tests. The order of exposition is a logical entry point, not an event before the universe or a privileged temporal beginning of the self-referential loop. A pair of mutually determining equations such as $x = (y + 2)/2$, $y = (x + 2)/2$ fixes $x = y = 2$ with no first value from which the other is manufactured; a calculator may approach the pair by iteration and someone may find it on a particular day, and neither belongs to the reason the answer is two.

The closure equation relates parts of one timeless object; it posits no backward signal in an external time coordinate. Observers recover the specification and later construct it in their own record order, and both stages belong to the same fixed point. In the terminating consensus theorems, repair descent supplies a well-founded order on the admitted updates for proving termination and comparing schedules; it is neither an observer's experienced time nor a universe-level clock.

What is reconstructed and what is constructed are two roles within this proposal. The observer who infers a law and the hardware that realizes it both belong to the candidate history. Finding that specification in a laboratory or checking it by computation occurs in the investigator's record order; neither supplies a temporal origin outside the history. If several specified closures satisfy the same tests, the results warrant that observational class without a uniqueness claim.

2.5 From closure to a testable specification

The executable observer gives the condition its finite meaning. A realization differs from a description because it carries the described dependencies under intervention: for a described update τ_a, realized operation F_a and fixed readout p, the one-step condition $p(F_a(s)) = \tau_a(p(s))$ extends by induction to every finite admitted sequence. A playback can match every frame of one history and fail this test as soon as a retained state is changed. The controls of Section 6 perform exactly this test on the running observer: disconnecting the detector or shuffling its bit with the same frequencies removes the feedback response, while the record-controlled execution keeps it.

Closure alone does not imply that a particular carrier, record law or initial state yields Standard Model physics. The executable-model hypothesis is the stronger claim that one simple, fully specified law does, and it is the form in which closure can fail. Its carrier architecture, refinement and composition, record instruments, memory law, scheduler, preparations and available controls are stated before testing. A failed specified law rejects that law and its tested interpretation; it reaches the wider proposal only when the rejected choice is proved necessary and exclusive. Changing the law, memory, state or decoder defines a new candidate.

2.6 Specification and theorem premises

The architecture, record instruments, memory law, scheduler, preparation services and allowed interventions form a declared specification. Observers may infer or design these ingredients; mathematics determines their consequences, and comparison determines whether the resulting process matches observed physics. Declaring a primitive fixes the model under study. Its consequences require proofs, and its physical adequacy requires tests; a specified input must remain distinguishable from a consequence derived from other inputs. The carrier, memory and control laws need not be uniquely selected by closure to be legitimate inputs. Their implementation and resource use must, however, belong to the process being studied.

Three premise families recur. A1 specifies the carrier, federation and response structure. A2 adds agreement on shared records and inner implementation of the complete response algebra. A3 adopts a least-informative state-selection principle on the stated feasible family. Closure alone proves neither the entropy minimizer nor the inner implementation assumption. Later theorems state which of these inputs they use and which additional hypotheses their conclusions require.

Quantum executions below use finite-dimensional complex Hilbert spaces, positive normalized density operators and completely positive instruments whose complete effects sum to the identity. These are explicit quantum premises, not consequences of classical normal-form uniqueness. A faithful reference is a strictly positive density operator $\rho > 0$; its modular flow is $\sigma_s(A) = \rho^{is} A \rho^{-is}$. The dimensionless parameter s requires a separate implemented calibration before it is read as duration. For fermionic fields the canonical anticommutation relations (CAR) are $\{c_i, c_j^\dagger\} = \delta_{ij} I$ and $\{c_i, c_j\} = 0$; the declared Fock representation fixes the exchange signs used later.

The physical interpretation also fixes a domain: one prepared history, a family of interventions, or every allowed input state. These are inequivalent requirements. Complete quantum instruments must retain the reference systems and all outcomes needed for the chosen domain; a record history need not assign simultaneous values to unperformed incompatible measurements. Likewise, a stationary probability law, an instantaneous repair fixed point and the closure of an entire history are distinct. Host runtime, repair counts, modular parameters and measured duration are related only by an implemented clock calibration. The operational observer below is defined by access and record interfaces, without an additional claim about consciousness.

3 A finite observer specification

The model rests on two objects: an operational observer and a twelve-port carrier with its native repair law. We define their interfaces and update rules before constructing the observer that runs on them.

3.1 Patches, records and repair

Definition 3.1 (Observer patch)**.** An observer patch is a finite object with local state, an observable boundary map, durable records, readback, and a set of authorized repair moves. A patch sees a fragment of the world: its own state and the boundary data of its authorized overlaps.

Definition 3.2 (Overlap and repair)**.** For patches x_i, x_j with an authorized overlap $e = (i, j)$, both induced boundary records must agree for the pair to hold public data. A repair move changes local state while preserving protected readout. A configuration is a normal form when no authorized repair applies.

An observer in this sense is defined by its access and record interfaces. It reads, writes and retains records, and it uses retained records to choose its next operation. Nothing in the definition refers to a background space, a global clock or a pre-assigned list of particles. Those enter later as readouts that observers can share.

3.2 The twelve-port carrier

Each local carrier has twelve boundary ports, thirty seams and twenty oriented triangular faces. The incidence has a selection theorem that uses no topological template, Euler count, group action or preassigned port count.

Theorem 3.3 (Icosahedral incidence from optimal readout)**.** *In a real three-dimensional response space, let unit vectors u_i read symmetric operators through* $\mathrm{tr}(P_i X)$*, $P_i = u_i u_i^{\mathsf{T}}$, with positive weights $\sum_i w_i = 1$, and put $M = \sum_i w_i |P_i\rangle\langle P_i|$. Informational completeness requires at least six distinct channels, and every choice obeys $\lambda_{\min}(M) \le 2/15$. With six channels, equality holds exactly for equal weights and the six icosahedral axes, up to an orthogonal frame. Exposing both signs gives twelve ports, positive overlap gives degree five and thirty edges, and the three-cliques give twenty faces. A deficit of at most 10^{-7} forces the same incidence, without forcing exact angles.*

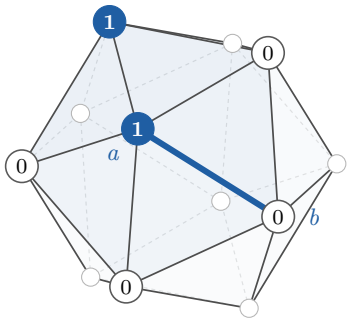
Proof. Sym(3) has dimension six and each term of M has rank one. Each $P_i - I/3$ has squared norm $2/3$, so the trace-free compression of M has trace $2/3$ on five dimensions, which gives the bound. Equality forces $M(X) = (2X + \mathrm{tr}(X)I)/15$. At six channels the matrix F with columns $\sqrt{w_i}P_i$ is invertible and $F^{\mathsf{T}}M^{-1}F = I$: its diagonal gives $w_i = 1/6$ and its off-diagonal gives $(u_i \cdot u_j)^2 = 1/5$. The Gram matrix $I + S/\sqrt{5}$ then has a sign matrix with $S^2 = 5I$, whose unique switching class fixes the geometry [7]. □

The response dimension, the quadratic readout, both objectives, inverse ports and positive-overlap edges are premises, and a physical metric requires operational reconstruction. Carriers federate through typed seam algebras and coherent triple overlaps into a nerve with a degree-one bridge to an oriented sphere. Native current records recover the local Gram; executed cross-carrier routing uses supplied cell centres and tangent frames. A1 in the theorem library declares this carrier, federation and response layer.

Two exact features of the carrier enter later constructions. First, the proper rotation group of the icosahedral packet is the alternating group A_5, with sixty rotations acting on the twelve ports, and the real port-coefficient space decomposes without multiplicity as

$$P_{12} \;\cong_{A_5}\; \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}. \tag{2}$$

On the six-dimensional golden sector the spectral projectors have entries in $\frac{1}{20}\mathbb{Z}[\varphi]$, $\varphi = (1 + \sqrt{5})/2$, each is the entrywise Galois conjugate of the other, and both triplets are irreducible real representations of real type; the projectors, characters and irreducibility certificates are machine-checked [55, 1]. Second, record keeping is integer valued: a write appends $+1$, a retraction appends -1, and readback sums atomic events per port. A conservative repair transfers one unit across a seam whenever the oriented mismatch has magnitude at least two. With $V(x) = \sum_i x_i^2$, each such repair strictly decreases V by $2(d-1)$ for mismatch $d \ge 2$, so repair terminates. The native repair law of Section 3.3 also moves load to the floor and ceiling of the endpoint mean, but at difference one it waits or swaps by its coin, so the native process keeps moving while its quadratic cost stays fixed.



twelve ports, thirty seams, twenty faces; a binary load of total two (filled ports)

$d = x_a - x_b, \qquad j = \frac{1}{2}\big[d - (2c-1)(|d| \bmod 2)\big]$

d	$c=0$	$c=1$	effect of the attempt
2	1	1	balance
1	1	0	swap or wait
0	0	0	no move
-1	0	-1	wait or swap
-2	-1	-1	balance

The move j carries j units from a to b, and the fair coin c reads no load. At $|d| = 2$ the attempt lowers $V = \sum_i x_i^2$; at $|d| = 1$ it keeps V fixed, so the native process keeps moving. The seam drawn has $d = 1$.

Figure 2: The twelve-port carrier and its native repair. Left: the icosahedral incidence of the twelve ports, with thirty seams and twenty faces, in an orthographic view with back edges dashed; the filled ports carry a binary load of total two, and the drawn seam (a, b) has $d = 1$. Right: the seam rule for every mismatch the total-two carrier admits.

3.3 The native repair law

The executable observer runs on one carrier with twelve nonnegative integer loads of total two. There are 78 such configurations, and the 66 binary configurations form an invariant shell. For an oriented seam (a, b) and a fair placement coin $c \in \{0, 1\}$ (Figure 2), the native repair law moves

$$d = x_a - x_b, \qquad j = \tfrac{1}{2}\big[d - (2c-1)(|d| \bmod 2)\big], \qquad y = x + j\,(e_b - e_a), \tag{3}$$

so load is conserved and the signed current j is the native observation. Conservation, one-step balance, a tie-break that reads no load and endpoint symmetry characterize this rule; on the total-two carrier they fix only the tie-break coin. Its quantum version is a record law, and seam locality, the classical law and the declared local number-phase and endpoint-swap covariance, a local phase symmetry of the repair channel, leave a convex family of them, each a table of five coherence multipliers [70]. The two laws used below have the same load update (3) and differ in which information they record. Write $p_e(x) = (x_a + x_b) \bmod 2$ for the parity of the seam's total load. Law C records the pair (c, j); law E records (c, j, p_e). Their Kraus operators are

$$L^C_{c,j} = \frac{1}{\sqrt{2}} \sum_{x:\, j(x,c)=j} |y(x,c)\rangle\langle x|, \tag{4}$$

$$L^E_{c,j,p} = \frac{1}{\sqrt{2}} \sum_{x:\, j(x,c)=j,\ p_e(x)=p} |y(x,c)\rangle\langle x|. \tag{5}$$

Each complete instrument preserves total probability. E exposes an extra parity record and therefore removes coherence between the corresponding parity classes when the record is ignored. It changes no classical load or current probability. C is the convention used for the primary construction; E provides a specified alternative and control. A current-only coherent implementation gives the most coherent member C_0; adding a coin readout gives C, and adding seam parity gives E. Classically forgetting a written coin is not the coherent C_0 implementation. E and the seam-total recorder are the two idempotent members of the classified family. Under a common definite-current record interface all members share every classical current statistic. One coherent interference experiment reads the multiplier g that separates C, E and the most coherent member (Section 13), and

nine source-executed measurements read every real coordinate of the table (Section 6). The unread channels have a source-independent spectral gap $(5-\sqrt{5})/60$ on the binary shell. Every admissible coherence table has the uniform binary state as its unique stationary state, and the rank-three load readings of Theorem 5.1 attain the slow rate; equilibrium therefore does not select C or E [70, 71]. Section 4.4 gives the entropy laws of the same repair.

The local operator $H_{\text{native}} = \sum_e (I - \Phi_e)$, with Φ_e an unread seam channel, is a positive Liouville operator on observables. It is not a Hamiltonian on carrier states: the executed channel attenuates coherence and cannot be unitary conjugation on this carrier. The separate field constructions must supply their own quantum evolution and physical energy interpretation [83, 69].

4 Public records and completed-record identities

The repair mathematics establishes when local agreement reaches global consistency and when nonzero cycle holonomy leaves a residue that local repair cannot remove. Two observers who share a boundary can disagree about what it shows. If repair stopped in different places depending on who acted first, the boundary would carry no fact of the matter. If it always stops in the same place, the final record is a public fact that no single observer owns. This section states when the second case holds, and then shows that completed records satisfy the familiar quantum identities on the declared complex algebra and state domain below.

A completed finite configuration x is a fixed point of the named repair-and-agreement map,

$$\mathcal{R}_{\text{cons}}(x) = x.$$

This map is distinct from the universe-level closure operator of Section 2. Fixed-point existence, observable determination and schedule independence are separate questions, and the following results isolate them [4, 6].

4.1 Observable normal forms

Definition 4.1 (Observable quotient system)**.** An observable quotient system is a tuple $\mathfrak{S} = (Q, C, \mathcal{B}, B)$, where Q is a set of configurations, $C \subseteq Q$ is the consistent subset, $\mathcal{B}$ is a set of protected records, and $B : Q \to \mathcal{B}$ is the record map. For $b \in \mathcal{B}$, write $C_b = C \cap B^{-1}(b)$.

The fiber C_b is unrealizable, reconstructing, or ambiguous according as its cardinality is zero, one, or greater than one. This trichotomy avoids assigning a state to a record that has either no consistent extension or several observationally indistinguishable extensions.

Theorem 4.2 (Canonical observable normalizer)**.** *The following statements are equivalent:*

(i) $B|_C$ *is injective;*

(ii) there is a unique map $\mathcal{N} : Q \to C \sqcup \{\bot\}$ *that preserves* B *on its* C*-valued outputs, fixes every point of* C*, is constant on each* B*-fiber, and returns* $\bot$ *exactly on fibers with no consistent extension;*

If $B(Q) \subseteq B(C)$*, the map is a total idempotent retraction* $Q \to C$*.*

Proof. Injectivity makes every nonempty C_b a singleton, which defines $\mathcal{N}$ and forces all its properties fiber by fiber. Conversely, if $c, c' \in C$ have $B(c) = B(c')$, the fixed-point and fiber clauses give $c = \mathcal{N}(c) = \mathcal{N}(c') = c'$. □

The normalizer is defined by observable fibers, independently of any repair schedule; a repair relation implements it when the relation preserves B, reaches the consistent set and satisfies a liveness condition. Approximate versions control the normalizer where exactness fails, through an optimal two-output distance bound and summable naturality defects along refinement towers. Canonicity carries no promise of cheapness: for succinct Boolean boundary systems, deciding whether a protected record has a consistent extension is NP-complete, and deciding injectivity of the record map is coNP-complete [4].

For the canonical single-site repair, every adaptive run is eventually constant because each nonstuttering attempt strictly lowers the mismatch count; pathwise weak fairness makes the stable state a normal form, and completeness with confluence sends every weak-fair run to the same public fixed point. These results select no scheduler and provide no stochastic hitting law, rate, physical clock, or refinement theorem. Attempts obey a sharp capacity law. A scheduler can delay the first genuine repair for any prescribed number of attempts. When every reducible record sees a genuine change among the next $q+1$ attempts, a bounded-waste upper horizon of $(q+1)$ times the initial mismatch count holds, and an independent-defect family attains the work-conserving case exactly at $q = 0$. The input-independent compiler produces a fixed federation with immutable input ports; genuine node repairs decrease a dependency rank, and pathwise weak fairness gives stable consensus and the correct output. Emission-order round robin finishes within its first n attempts, and the single-consumer structure bounds every accepted path by $n(n+1)/2$, with an emitted family attaining $\Theta(n^2)$. General well-formed circuits instead admit $2^n - 1$ accepted steps, attained by a fanout chain, where n counts the emitted nodes. These statements concern canonical node repairs: the weak patch-frame relation can stutter fairly forever, and no physical scheduler, clock, energy or hardware cost follows from these operation counts [4, 6].

4.2 Public records from local repair

For a prepared batch of support-closed transactions, join τ and σ in the conflict graph when $W_\tau \cap (R_\sigma \cup W_\sigma) \neq \varnothing$ or $W_\sigma \cap (R_\tau \cup W_\tau) \neq \varnothing$, and iterate to a fixed point: compute semantic support inside every aggregate, rebuild the conflict graph on the expanded read and write sets, merge every connected component, and repeat until the partition and all aggregate supports are unchanged.

Proposition 4.3 (Transactional local diamond)**.** *Let an accepted repair be an aggregate transaction with read set R_τ, write set W_τ, a read snapshot, and a snapshot-determined payload. Assume:*

(i) every read set contains the semantic dependency closure of its write set, and every affected acceptance functional is revalidated at commit;

(ii) the terminal closure above exists, and each of its connected conflict components has one coherent canonical aggregate payload;

(iii) a prepared component whose snapshot remains valid survives commits of independent components; and

(iv) accepted commits preserve the protected record and strictly decrease the integer descent functional.

Then every one-step quotient peak $t \longleftarrow s \longrightarrow u$ has a quotient state v with $t \longrightarrow v \longleftarrow u$.

Proof. Two different first steps cannot originate in one conflict component, since that component has a unique aggregate payload, so they originate in distinct components of the terminal closure.

Rebuilding the graph after every aggregate support expansion makes their final read and write sets disjoint, so neither commit changes the other's read snapshot, and semantic dependency closure ensures neither changes an acceptance functional consumed by the other. Revalidation and the surviving component rule make both second commits legal, and with snapshot-determined payloads and disjoint writes the two application orders agree; protected-record preservation carries the equality to the physical quotient. □

Theorem 4.4 (Consensus normal form)**.** *Suppose the accepted repair relation satisfies the hypotheses of Proposition 4.3, so that it has the one-step diamond property, strictly decreases the quadratic descent functional V of Section 3, and is repair complete, meaning that its normal forms are exactly the consistent states. Every maximal asynchronous repair schedule from one initial state then terminates at the same consistent quotient normal form. The induced global repair map is idempotent, fixes consistent states, and preserves the protected record.*

Proof. Quadratic descent makes the accepted relation terminating. Proposition 4.3 makes it locally confluent, so Newman's lemma [48] makes it confluent. A terminating confluent relation has a unique normal form below each source, and repair completeness puts that normal form in the consistent set. Applying the normalizer a second time does nothing, and protected-record preservation holds along every accepted edge. □

Atomic commits, disjoint write sets, or termination alone do not imply the local diamond; countermodels omitting semantic dependency closure, canonical aggregation or the surviving component rule are retained with the finite receipt. This is the precise sense in which a public fact is what survives every authorized comparison: the public object is the protected quotient record, independent of the order in which observers repaired.

Repair also has a residue that it cannot remove. On a fixed graph with positive mismatch weights and fixed edge offsets, nonzero cycle holonomy forces a strictly positive minimum mismatch, and no repair of the vertex assignments eliminates it while the offsets are fixed [6]. The matter hypothesis uses this frustration as a candidate mechanism (Section 8.4).

The law runs at scale. The public run uses the level-six cell rung of the geodesic icosahedral refinement tower: 81,920 exact twelve-port carriers with one declared gluing of ports to neighboring cells, 2,580,480 seams in all, and the seam-mean retraction as repair primitive. Sixteen shuffled schedules from one seeded load terminate at the balanced class with no accepted move raising V, and all sixteen return one quotient hash equal to the component multiset fixed by the initial totals. A standalone verifier that imports no simulator code rebuilds the component multisets from the primitive arrays, replays one schedule from its seed and recomputes every hash [2]. The level-ten rung, with 20,971,520 carriers and 660,602,880 seams, terminates all sixteen schedules at the balanced class in 84 to 100 sweeps with one quotient hash fixed by the loads. The excess-energy contraction per sweep lies between 0.81 and 0.83 across rungs five to nine, and these finite readings prove no all-size mixing or termination-time bound. Stationary continuation adds a constructive fluctuation result: for fair integer repair on a connected weighted graph with independent edge clocks of total rate one the fixed-occupancy law is reversible, centered loads have lagged covariance $\kappa e^{-|t|L_G/2}\Pi$ in the edge-clock time t, with L_G the weighted graph Laplacian, $\kappa = r(p-r)/(p(p-1))$ for r raised occupancies on p ports and Π the projection off constants, and fixed linear readouts obey a martingale central limit theorem [7].

The same records let observers recover the specification itself. At carrier scale, recovery from event logs under the declared probe and operation grammar returns the twelve ports, thirty seams, inverse pairing, automorphism and rotation groups of orders 120 and 60, mean rule, schedule, rank-three Gram and Lie split $1+3+8$. Re-instantiation preserves the invariant vector for two iterations,

and tetrahedral and octahedral controls recover their distinct specifications [2]. Complete federation gluing is not identified by the retained routing logs: within the declared port-gluing interface a two-edge switch preserves every carrier degree, occupied port and source traversal while changing the carrier graph's triangle count from 14,000 to 13,990, and native executions reproduce identical bytes for both. The alternative is not asserted to belong to the canonical geometric family. Full-gluing recovery requires further observations or a justified restriction of the candidate class.

4.3 Completed records and quantum identities

Once a compare, write and verify slice has completed, its accessible events are represented in a finite-dimensional complex C^*-algebra with a positive normalized state and completely positive instruments. This algebra-state representation is an explicit input; the consensus theorem identifies which record is public and does not by itself select complex quantum theory over every alternative probability model.

Theorem 4.5 (Finite public-record identities)**.** *Let P_E be the projector of a completed public event E and ρ the normalized state of the completed record surface. The event weight and nonzero-weight update*

$$\Pr(E) = \operatorname{Tr}(\rho P_E), \qquad \rho \,|\, E = \frac{P_E\, \rho\, P_E}{\operatorname{Tr}(\rho P_E)} \tag{6}$$

define a normalized probability measure on every orthogonal event partition and a normalized positive state on the post-event corner. For self-adjoint dichotomic observables A_0, A_1 and B_0, B_1 in commuting record subalgebras, with $A_i^2 = B_j^2 = I$, the combination $S_{\mathrm{CHSH}} = \operatorname{Tr}[\rho\{A_0(B_0 + B_1) + A_1(B_0 - B_1)\}]$ obeys $|S_{\mathrm{CHSH}}| \le 2\sqrt{2}$ [29, 30]. The bound is attained exactly on the committed slot-split interface by the maximally entangled state with Pauli and quarter-rotated projections, while every state diagonal in the record basis with diagonal observables valued in $[-1, 1]$ obeys $|S_{\mathrm{CHSH}}| \le 2$.

Proof. Positivity of ρ and $0 \le P_E \le I$ give $0 \le \operatorname{Tr}(\rho P_E) \le 1$, and $\sum_k P_k = I$ normalizes every partition; for nonzero weight $P_E \rho P_E / \operatorname{Tr}(\rho P_E)$ is positive and normalized. For $\mathcal{B} = A_0(B_0 + B_1) + A_1(B_0 - B_1)$ the commuting-subalgebra calculation gives $\mathcal{B}^2 = 4I - [A_0, A_1][B_0, B_1]$ [31], so $\|\mathcal{B}\| \le 2\sqrt{2}$. The saturating witness and the diagonal bound are exact finite calculations [5]. □

At the level of weights the Born form is derived. Every valuation on the effects of the finite surface that is nonnegative, normalized and additive on coexisting effects equals $E \mapsto \operatorname{Tr}(\rho E)$ for a unique density operator ρ, in every finite dimension including two, where the projector-only Gleason theorem fails, and the derivation consumes no continuity axiom [27, 28]. The projectors alone do not select the update: an explicit swap-twisted instrument has the same effects and different updates. For a complete Kraus instrument, basis survival and one coherent survival probe per nontrivial outcome block select both the effects and the Lüders map. Approximate tests bound complete outputs with arbitrary references; trusted preparation and readback are premises [34, 5]. The same surface carries the copying obstruction: if one ancilla-free isometry copies two sharp states from a common blank, preservation of inner products forces their overlap z to satisfy $z^2 = z$, so two distinct alternatives copied by one device are orthogonal [32, 33].

Native repair histories construct a functional of this form on a full matrix algebra. The complete-attempt transfer P of the 66-state carrier comes from all 3960 endpoint branches and obeys $2I/3 \le P \le I$. The twelve occupation fields and connected transfer paths generate all 4356 matrix units, scaled closed-history counts determine their moments, and the unique positive density is $\rho_L = P^L / \operatorname{Tr} P^L$. Its canonical complex-linear extension is additive and positive, so the Born theorem applies on this reconstructed algebra. A current-only return census implements

the required configuration comparison, with at most 11398 attempts per experiment at the stated settings and systematic error at most 3.76596×10^{-6} per reconstructed effect statistic. Finite polynomials reconstruct ordered modular correlations with the common generator $-\log P$, with bias below 4.302×10^{-10} in the displayed three-time example. The entries of this functional are classical endpoint probabilities divided by the return probability, every connected positive-definite symmetric stochastic matrix gives the same result, and native C and E evolution converges to $I/66$ without preparing it. Direct native acceptance effects are diagonal. Complex tomography uses supplied coherent services, and the operational bridge requires equal-effect probability independence and disjoint common implementations of coexistent pairs [5, 8].

4.4 Entropy and thermodynamic laws from repair

For a conditional-resampling repair law with a stated reference and energy calibration, finite theorems give the second law, a Clausius inequality, fluctuation relations and a Landauer bound [8]. Native repair realizes this structure directly. On binary populations, averaging the two fair-coin outcomes of a seam gives $P_e = (I + \mathrm{swap}_e)/2$, exactly uniform resampling conditional on the outside-seam loads. The connected seam ensemble has one uniform stationary reference, which uniquely maximizes entropy in its sector, and

$$S(pP_e) - S(p) = D(p \| pP_e) \geq 0$$

for the unread population; fixed seam words telescope this identity. Retaining the complete endpoint and coin/current tape gives $H(X_n, J) = H(X_0) + n \log 2$, so the marginal entropy increase loses nothing from the complete record. The modular energy of this branch is constant, which leaves its temperature undefined.

A native memory refresh supplies a nonconstant calibration. A stopped coin-prefix program records 0, 10 or 11 after one, two or two attempts, with weights $(1/2, 1/4, 1/4)$ that ignore the hidden configuration. The stored-prefix cost $E = n \log 2$ gives weights e^{-E}, and overwriting the memory with a fresh prefix is exactly its global heat bath in at most two native attempts. Agreement of b independent prefixes gives the Gibbs family of the same cost at integer multiplier b. Keeping the old memory after disagreement gives a lazy reversible kernel with that stationary reference, bounded retries keep every rejection outcome with an explicit geometric error, and almost-sure retry gives the exact heat bath at finite expected cost. A Lean instance constructs the repair and calibration data from these counts and applies the finite four-law core, with gap $\log 2$ and a unique ground label. That core holds for every full-support law with $E = -\log(\mathrm{ref})$ and formalizes no contact relation, the fair coin is a declared input, and record cost is distinct from laboratory heat, temperature and total implementation work [8].

Pressure needs a strain law. The family $H_\alpha(V) = Q(V/V_0)^{-\alpha}$ has the same fixed-volume energy histories and $p/\rho = \alpha$, so consensus about identical fixed-volume records cannot select the strain derivative. Three geometrically spaced energy readings identify it within the held-state family $C + A(V/V_0)^{-\alpha}$, with exact intervals propagating meter error, and a complete two-attempt native E census reads independently calibrated charge-diagonal forces. An additive energy offset changes p/ρ while leaving pressure unchanged. Volume and energy calibration are supplied [8].

5 Three spatial directions

The carrier incidence fixes the dimension of the carrier's own position readback: the slow band is the three-dimensional irreducible representation of the icosahedral group, so the number three is a property of the committed carrier.

Theorem 5.1 (Intrinsic three-dimensional carrier completion)**.** *Let A be the twelve-port adjacency, let $M = I - (5I - A)/60$ be the uniform scalar repair mean, and take the complete equally weighted centered probe census. The normalized infinite-response kernel converges to the intrinsic Gram $G = 4P_{\mathrm{slow}}$, of rank three. The twelve integer port loads have antipodal-odd quotient $\mathbb{Z}^{12}/\{x : x_p = x_{-p}\} \cong \mathbb{Z}^6$, and the thirty-seam boundary has image exactly the even-sum sublattice D_6, with one residual parity coset. Under the pullback of the response-selected Gram metric, both integer modules embed densely in one abstract continuous three-dimensional Euclidean carrier. The sixty proper carrier maps act faithfully and isometrically on the rank-three range and extend to that completion, and cumulative records act by exact internal translations.*

The mechanism is spectral. The icosahedral adjacency has eigenvalues 5, $\sqrt{5}$, -1 and $-\sqrt{5}$ with multiplicities 1, 3, 5 and 3, so the repair mean M has eigenvalue $1 - (5 - \sqrt{5})/60$ on a three-dimensional eigenspace and smaller eigenvalues on everything else nonconstant. Repeated local averaging forgets every nonuniform pattern except a three-dimensional one, and that surviving pattern is the triplet of (2) carried by the port position vectors. The same number $(5 - \sqrt{5})/60$ reappears as the quantum spectral gap of the native repair channel (Section 3.3).

The completion is taken after slow-response normalization and the infinite-step limit; at finite step count the kernel has rank eleven on the centered space and rank six on the signed antipodal sector, so the order of limits matters. The metric is the response-selected Gram pullback, distinct from the usual six-dimensional lattice metric on D_6. Thirteen finite covariance and isometry table identities are discharged in Lean with `native_decide`, so those subreceipts trust the native compiler in addition to the kernel and are not kernel-only proofs. Displacement is a record difference and cumulative records translate the completion, so position is the carrier's own readback of its accumulated records. In the executable observer the same rank-three readout attaches to all joint histories of the interacting process, and a binary controller extension executes displacements spanning three independent directions [69]. Reading the completion as physical position consumes the realization map (M1) of Section 12.3 [1, 7]. The rank-three readout belongs to one carrier's response. When carriers join into the refining spherical screen of Section 7.1, the continuum form of their repair history has spectral dimension two, the dimension of the screen, and bounded-energy families lose every within-carrier component, this band included, at the present full-port energy scaling; a three-dimensional bulk reading of the screen requires observation maps that the native energy does not supply.

6 The source specification in operation

The finite theory has to run as a process that observers can access. Its specified carrier, complete instruments, scheduler, memory, controls and preparations define that process. Source executions check that the same operations produce the claimed conditional records, including the outcomes that an observer does not see. The construction paper gives the full implementation and certificates [68, 69].

Records affect later operations. A two-event observer writes the first seam current, selects its next seam from that record and keeps the second outcome. Exact enumeration gives final port-zero occupation 35/264, versus 1/6 for a disconnected or frequency-matched shuffled selector. The comparison isolates the causal role of the retained record within this specified rule. It does not select that rule as the one realized by nature [72].

Observers share one source. Two observers with overlapping port access read a continuing native history, each with its own incident-event clock. Their decoded values at the common port agree on all 5,768 retained output words, including waits and timeouts; a separate transfer recursion reproduces the same outputs exactly. Federated-screen executions and a locality estimate carry complete root-observer outputs across declared larger regions [78, 84]. These are finite agreement and locality results. Physical position and duration require additional realization maps.

Quantum instruments and memory. The C and E instruments implement the same classical seam update while recording different information about a quantum input. Interference can distinguish them, and a finite measurement menu identifies the coordinates of their declared record-law family [82]. A fixed 570-attempt native preparation accepts arbitrary total-two inputs, including entangled references, with error below 3.72×10^{-8}. Ignoring its warm-up tape adds that error once to all 8,195 records of the capped phase/work continuation; coherent-compiler error is retained separately [8]. On its supplied work register, gauge flux, a charged pair and scalar sites have noncommuting evolution and record feedback. A smooth four-read continuation has complete $O(1/d)$ error, and active recovery protects a specified logical qubit [73, 74, 75, 77].

The source executions establish that declared operations compose and that records can control later field and repair actions. They do not derive an autonomous observer population, a physical clock or an interacting relativistic continuum from the finite tests. The physical status of those joins is stated in Section 12.

6.1 Measurement and public records

A private outcome becomes public when authorized comparisons preserve it through repair to a common normal form. Event projectors, weights and conditioning then act on the completed algebra-state surface of Section 4.3. The running specification demonstrates this transition with full conditional instruments and untouched reference systems. Quantum identities use the supplied complex algebra-state representation; the finite measurement protocols verify its declared implementation on their stated domains [68, 76].

7 Fields, clocks and causal order

The same carrier selects its field action. Write D and C for the seam-by-port gradient and face-by-seam curvature, so that $CD = 0$.

Theorem 7.1 (Local Maxwell selection)**.** *A real quadratic seam action $S(A) = \text{const} + \ell^{\mathsf{T}} A + \frac{1}{2} A^{\mathsf{T}} H A$ that is gauge invariant, has off-diagonal Hessian entries only between seams of one face, and is covariant under the sixty carrier rotations is exactly $S = \text{const} + \frac{\kappa}{2}\|CA\|^2$. For positive weights ϵ, β, a real linear autonomous evolution with $\dot{B} = -CE$, conserved $(\epsilon\|E\|^2 + \beta\|B\|^2)/2$ and reversal with E even and B odd is exactly $\dot{E} = (\beta/\epsilon) C^{\mathsf{T}} B$ on closed flux.*

Proof. Gauge invariance is $HD = 0$ and $D^{\mathsf{T}}\ell = 0$. At an edge of a face and its opposite vertex, locality permits exactly the other two edges in that entry of HD, and the signs of $CD = 0$ propagate one weight w_f over the face. Subtracting $\sum_f w_f C_f^{\mathsf{T}} C_f$ leaves a diagonal matrix annihilating D, hence zero. Transitivity on faces equalizes the weights, and the sum of the sixty signed seam actions vanishes, which removes ℓ. For evolution, conservation makes the electric block skew and fixes the cross block on closed flux, and reversal removes the skew block [7]. □

Exact rank computations and countercontrols verify the classification. The same form is the leading response of a smooth conservative field law, with a uniform finite-time bound $\|x(t) - e^{tG}x_0\| \le (L/2)J_0 r^2|t|$ and no linear ansatz; exact nonlinear Gauss conservation needs an additional condition. On a supplied Euclidean domain, the complete equal-weight seam symbol Λ_a obeys $0 \le r^2 - \Lambda_a(k) \le a^2 r^4/20$ at $r = |k|$, and its reversible curl-pair group converges to the local Maxwell group with $\|U_a(t,k) - U_0(t,k)\| \le \min\{2, |t|a^2 r^3/20\}$. The premises, field identification, coefficients and clock are inputs, and native repair does not select them.

The action principle itself has a source form.

Theorem 7.2 (Derived action of a source chain)**.** *Let P be a strictly positive row-stochastic kernel on a finite state set Ω and π a strictly positive initial law. On paths with n transitions put $S_{\log}(\gamma) = -\sum_i \log P(\gamma_i, \gamma_{i+1})$ and $R(\gamma) = \pi(\gamma_0)|\Omega|^{-n}$. The Markov path law is the exponential tilt of R by $S_{\log}$ at multiplier one, with partition constant $|\Omega|^{-n}$, and a pair (S', λ') reproduces it exactly when $\lambda' S' = S_{\log} + c$ for a constant c.*

Exponentiating the negated action recovers the transition product, and row normalization telescopes the partition sum; the uniqueness clause follows by comparing logarithms at a common path, and the statements are checked in Lean [8]. The committed source chain instantiates the theorem exactly. Its bare repair count reproduces the path law at no multiplier, which makes the log-transition action the derived object. Physical field actions are a further attachment.

7.1 The native screen operator and a conditional spacetime field

On the geodesic screen of level l, let $V_l = 240 \cdot 4^l$ and prepare the uniform canonical state with total load $V_l/3$. Independent uniform seam selection and fair native repair define one complete occupation history. Write $\chi_l = q_l(V_l - q_l)/[V_l(V_l - 1)]$ for $q_l = V_l/3$.

Proposition 7.3 (Native continuum operator)**.** *Along a common refinement subsequence, the native cell Green operators converge in norm to a compact positive injective operator G on the mean-zero observation space. After restoring the constant mode, its inverse A generates a regular strongly local conservative form, with domain H^1, and*

$$n/49920 \le \lambda_n(A) \le 3n/2.$$

The complete density readouts

$$Y_l(\tau; f) = \frac{\sum_p f_l(p)[n_{\lfloor 1260\, 16^l \tau \rfloor}(p) - 1/3]}{\sqrt{V_l \chi_l}}$$

converge jointly at finite sets of times to a stationary reflection-positive Gaussian field with covariance $\langle f, e^{-|\tau - u|A} g\rangle$. The admitted coherent continuation converges strongly to $\cos(t\sqrt{A})$. With the constant mode restored,

$$\|\mathbf{1}_F e^{-\tau(A \oplus 0)} \mathbf{1}_E\| \le e^{-d(E,F)^2/(2\tau)},$$

and the cosine propagator vanishes for $|t| < \sqrt{2}\, d(E, F)$.

Proof sketch. The native cell Poincaré bound $\|(I - P_k)u\|^2 \le 624\, 4^{-k} \langle u, B_l u\rangle$ gives uniform compactness of the Green operators and norm convergence. Mosco convergence passes their local Markov forms to the limit; min–max estimates give the displayed spectrum. Conditional finite-population limits apply to the actual identity-or-swap repairs. The Gaussian heat bound gives finite propagation of the admitted cosine family [95, 96, 97, 85, 86]. □

Thus one source operator controls static covariance, relaxation and a chosen coherent wave service. Its spectral dimension is two. The $\lambda_n \asymp n$ count excludes identifying it, up to scale and bounded mass shift, with a three-dimensional elliptic bulk operator or a first-order boundary operator. A physical bulk interpretation needs an additional radial observation map. Passive repair supplies no wave sector; its attempt count and the coherent wave parameter require clock calibration. The native stationary reference supplies no physical quantum vacuum or universal area-entropy coefficient [68, 87]. On the binary repair sector its complete instrument has unique stationary state $I_{66}/66$. A declared chart readout gives faithful three-dimensional references conditional on individual native current records; averaging those records gives $I_3/3$. Recorded moves have mean $5/33$ per attempt and asymptotic variance $1597/11979$, including temporal correlations. These are conditional references and an attempt-count clock, without a physical energy calibration [8].

A specified radial interval and field service give the product covariance $(-\partial_r^2 + A \oplus 0)^{-1}$. If its hyperbolic coefficient density is $C = \mu \operatorname{diag}(-1, a)$, the effective metric is

$$g = \sqrt{-\det C}\, C^{-1} = \mu\sqrt{\det a}\, \operatorname{diag}(-1, a^{-1}).$$

The spatial coefficients are read from the source; the radial architecture, hyperbolic service and preparation are explicit design inputs.

Proposition 7.4 (Smooth effective field descriptions)**.** *Smoothing the native angular coefficients while retaining the radial Dirichlet interval gives norm-resolvent convergence, with gap $(\pi/L)^2$. For every fixed finite admissible menu of observer readouts on bounded wave times and every $\varepsilon > 0$, a smooth static $(1+3)$-dimensional Lorentzian metric carries a free scalar field whose complete predictions agree within ε, in addition to the stated source and clock errors.*

The proof transports the changing measures into one Hilbert space and combines compact Mosco convergence with continuity of the covariance, response and finite-history operators [68]. It proves a common finite-menu approximation, without selecting a radial architecture, providing a smoothing rate, or producing a responsive metric. The three-dimensional fermion limits in Section 9 have their own source and refinement hypotheses.

7.2 A causal precursor from records

The oriented conformal spherical support carries the classical isomorphisms

$$\operatorname{Conf}^+(S^2) \;\cong\; \operatorname{PSL}(2,\mathbb{C}) \;\cong\; \operatorname{SO}^+(3,1), \qquad H^3 \;\cong\; \operatorname{SO}^+(3,1)/\operatorname{SO}(3), \tag{7}$$

which identify the three-dimensional space of future unit timelike directions [54]. Authenticated records separately supply events and their order. Every quotient-visible register version carries the identifier of its last semantic writer, each commit certifies the versions its acceptance depends on, and the informational order is the transitive closure of authenticated read-from parenthood, an authenticated analogue of happened-before [44]. As an abstract finite poset it satisfies the order-theoretic causal-set axioms, and every finite causal set is the generated order of some authenticated log [1]. Twelve declared null-direction tests certify the conformal Lorentz class of a supplied invertible chart map. The actual single-queue controller instead has a chain of full storage dependencies; identifying physical events and matching their clock and cone to the field are separate source obligations [8, 9].

Theorem 7.5 (Source-derived finite (1+3) causal precursor)**.** *Let $\mathcal{L}$ be an authenticated finite semantic-event log with generated strict order $\prec_{\mathcal{L}}$ and canonical source height $h_{\mathcal{L}}$, zero on roots and*

one plus the maximum parent height otherwise. Then $h_{\mathcal{L}}$ strictly increases along $\prec_{\mathcal{L}}$ and equals the maximum length of an authenticated parent chain ending at each event. Let V_{src} be the rank-three source Gram quotient with positive form g_{src}, and put $W_{\mathrm{src}} = \mathbb{R} \oplus V_{\mathrm{src}}$ with $Q(t, x) = t^2 - g_{\mathrm{src}}(x, x)$.

(F1) W_{src} has dimension $1 + 3 = 4$ and Q has one positive and three negative directions;

(F2) source-unit directions, S^2 and future-null rays are exactly equivalent;

(F3) a placement $F(e) = (\tau h_{\mathcal{L}}(e), x(e))$ sends every generated precedence into the future cone whenever each parent edge obeys $g_{\mathrm{src}}(x(f) - x(e), x(f) - x(e)) \le \tau^2 (h_{\mathcal{L}}(f) - h_{\mathcal{L}}(e))^2$;

(F4) every finite log admits an injective placement of this form that sends generated ancestry into the future cone and makes distinct same-height events spacelike; and

(F5) for a placement obeying the edge condition of (F3), if the spatial readback separates each equal-height layer and every unrelated pair with $h_{\mathcal{L}}(e) < h_{\mathcal{L}}(f)$ is spacelike, then $e \preceq_{\mathcal{L}} f$ holds exactly when $F(f) - F(e)$ is future causal, so F is an exact order embedding preserving every causal interval.

Proof. Dimension is additive and positivity of g_{src} gives the inertia; induction over parents bounds every ending chain by the height and constructs an attaining chain; the edge inequality and transitivity of cone and ancestry give (F3); an enumeration along one axis with $\tau = n + 1$, n the number of events, gives (F4); and the two geometric checks exclude every unsupported cone comparison in (F5). The finite statements are formalized [1, 9]. □

Read informally, the theorem combines two things observers have. Who read whose write is a partial order, and the longest chain of reads ending at an event gives it a height. The rank-three carrier gives three spatial directions and, with a real axis, a light cone. An event placement that respects both turns the provenance order into a causal order in a four-dimensional Lorentzian carrier.

The first two clauses construct the carrier and signature from an independent real axis and the rank-three source carrier, without assuming a rank-four chart; source height enters only through placements, and the scaled height is no operational clock. A physical manifold requires a source-causal continuum certificate: one refinement family supplying physical event identification, source-selected order and cone embeddings, refinement maps, a calibrated count density, agreement across independent dimension and manifoldlikeness tests against Poisson-sprinkled controls, and convergence to a distinguishing time-oriented Lorentzian geometry [35, 36, 37].

Proposition 7.6 (Composition of the continuum certificate)**.** *For a source-causal continuum certificate, causal order fixes the conformal Lorentzian geometry of the admitted distinguishing limit and the calibrated count measure fixes its conformal scale [42, 43]; the local model is $(1 + 3)$, so a limit satisfying the dimension and topology clauses of the certificate is an effective four-dimensional time-oriented Lorentzian event manifold. If on the same family a causal-set d'Alembertian or scalar-curvature estimator converges as an independent curvature diagnostic [40, 41], an explicit tensor-curvature reconstruction converges to the Einstein tensor or the small-ball null-balance route identifies the limiting geometry tensor with G_{ab}, and stress and Ward–Bianchi data converge with calibrated coupling, then the smooth limit is $G_{ab} + \Lambda g_{ab} = 8\pi G\, T_{ab}$.*

A source-record population realizes the flat case.

Theorem 7.7 (A source-record causal and count limit)**.** *In the rank-three conservative repair carrier, let* $\varphi = (1+\sqrt{5})/2$, $L = 2/\sqrt{\varphi+2}$ *and* $q = F_n$ *the* n*-th Fibonacci number,* $n \geq 4$*. The population* $S_q = L\{b\varphi - \lfloor b\varphi \rfloor : 0 \leq b < q\}^3 \subset [0,L]^3$ *consists of* q^3 *distinct conservative source-record images. Retain it at each layer and let every next-layer event read all preceding-layer sites within* $a_q = L/\sqrt{q}$*, with model duration* $\Delta_q = a_q/c$ *for a declared speed* c*. As* $q \to \infty$ *the order converges away from null separation to the speed-*c *causal order, and with* $\rho_q = q^3/(L^3\Delta_q)$*, for fixed interior timelike diamonds,*

$$\frac{|I_q|}{\rho_q} \longrightarrow \mathrm{Vol}(I), \qquad \frac{2C_q}{|I_q|(|I_q|-1)} \longrightarrow \frac{1}{10},$$

where C_q *counts strict causally ordered pairs; for vertical duration* T*,* $\mathrm{Vol}(I) = \pi c^3 T^4/24$*.*

Corollary 7.8 (Proper-time ratios from record counts)**.** *If* I_q, J_q *approach interior timelike diamonds of proper durations* $\tau_I, \tau_J > 0$*, the observer readout* $R_q(I;J) = (|I_q|/|J_q|)^{1/4} \to \tau_I/\tau_J$*. Within positive volume-only readouts additive on collinear inertial intervals, the fourth root is unique up to units.*

At finite regulator the population is produced by the carriers themselves: each holds its record as port loads, reads its position through the rank-three response and decides its neighbors from that readback, and the precedence generated from their authenticated reads equals the layered order through $q = 89$, that is 704,969 carriers reading one another over ten rounds. On the largest interior diamond the inverted Myrheim–Meyer dimension reads 2.83, 4.38, 3.55, 3.84 and 3.92 at $q = 13, 21, 34, 55, 89$ [38, 39]; the readings at $q = 5$ and $q = 8$ lie below the regime of the theorem. The ordering fraction of the central diamond reads 0.088, 0.138, 0.092, 0.093 and 0.094 against the proved flat value 1/10, and the count clock reads 1.71, 2.09, 1.91 and 2.05 at $q = 13, 34, 55, 89$ against the model-time ratio 2. Covariance requirements select the cone: a closed convex pointed displacement cone preserved by the irreducible icosahedral rotations and by boosts along one axis is the future or past cone of $t^2 - \|x\|^2$. The read law and model clock are declared, and the ordinal height of a history with self-waits is no proper-time instrument [9].

The executable carrier has its own finite causal bound. For a fixed seam schedule and shared local coins, dependence crosses at most one seam per ordered repair event, and retained signal witnesses attain the bound; this is a speed in seams per native event. Remote repairs can carry a difference to a receiver before its first local tick, so the bound is not one seam per receiver tick [88, 69]. Complete endpoint records give a forward-cone placement in the registered response geometry, and a four-event native execution has a Lean instance proving that implication and the failure of its converse [1].

8 Gauge structure, particle operators and the pixel scale

The finite theorem fixes a current algebra under explicit response premises. A chosen carrier, memory law and field service then realize its currents and mass operators. These are different steps of one construction: the Lie type constrains admissible interactions, while the full executable specification fixes their coefficients. Recovering or choosing that specification is allowed; its agreement with physical particle records is an additional requirement.

8.1 The finite gauge theorem and matter representation

Let A be the icosahedral adjacency matrix and J its antipodal involution. Exact incidence gives

$$10J = A^3 - 4A^2 - 5A + 10I, \qquad P_{12} = \mathbf{1} \oplus \mathbf{5} \oplus \mathbf{3} \oplus \mathbf{3}'. \tag{8}$$

The even port coordinates carry $\mathbf{1} \oplus \mathbf{5}$; the odd coordinates carry the two triplets. The response premise of Section 2.2 requires an injective derivative $D : P_{12} \to \mathfrak{u}(H)$, a commutator-closed image with positive trace pairing, and internal implementation of every proper carrier rotation. Internal implementation means that its projective implementer factors into the connected response group and a pointwise centralizer. This is the response-endogeneity clause identified as A2 in the library; it is not implied by the port count alone [99, 11].

Theorem 8.1 (Gauge Lie-type forcing)**.**
Under these complete-response and internal-implementation premises,

$$\mathfrak{g} = D(P_{12}) \cong \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3).$$

Its center is the uniform port line. The three-dimensional simple ideal carries one triplet, and the eight-dimensional ideal carries the other triplet and the five-dimensional band.

Proof. The trace pairing makes the twelve-dimensional image compact. Internal implementation makes the A_5 action inner, so it fixes the center pointwise. The transitive port action has a one-dimensional fixed space, hence the center has dimension at most one. With zero center, the only compact semisimple type of dimension twelve is $\mathfrak{su}(2)^4$. Inner automorphisms preserve its simple ideals, each of which has either zero or three A_5-fixed directions, contradicting the one-dimensional fixed space. The center therefore has dimension one; the unique compact semisimple type of dimension eleven is $\mathfrak{su}(2) \oplus \mathfrak{su}(3)$ [65]. □

The premises have an explicit finite matrix realization. Six antipodal axis vectors form a matrix U over $\mathbb{Q}(\sqrt{5})$; $\sigma(U)$ is its Galois conjugate. For even coordinates b, odd coordinates d, and $c = \frac{1}{6}\sum_i b_i$, put $\Phi_0(b - c\mathbf{1}) = \sum_i (b_i - c) u_i u_i^{\mathsf{T}}$. For nonzero real band coefficients,

$$K(f) = \operatorname{diag}\Big(\lambda_3 \widehat{Ud} + i[\lambda_1 c I_3 + \lambda_5 \Phi_0(b - c\mathbf{1})], \lambda_{3'} \widehat{\sigma(U)d}\Big), \qquad \widehat{x}y = x \times y. \tag{9}$$

Proposition 8.2 (Finite response realization)**.** *The map K is injective and equivariant, its image is commutator-closed, and proper carrier rotations act internally. Its pulled-back bracket has*

$$(P_{12}, [\ ,\ ]_K) \cong \mathfrak{u}(3) \oplus \mathfrak{so}(3). \tag{10}$$

The six axis projectors span the symmetric matrices, while the triplets span the skew matrices in the two blocks. These rank and equivariance identities are exact over the full sixty-element rotation group. The representation and nonzero coefficients are design inputs. A zero bracket on the same port module violates the internal-implementation premise; it is not excluded by geometry alone. The gauge specialist gives the complete finite certificate and the boundary between analytic classification and machine-checked algebra [11].

Theorem 8.3 (Finite matter image)**.** *For the supplied trace-balanced module $V = \mathbb{C}^3_{-1/3} \oplus \mathbb{C}^2_{1/2}$,*

$$\Lambda^2 V \oplus \Lambda^4 V = Q \oplus u^c \oplus e^c \oplus d^c \oplus L, \tag{11}$$

where $Q = (3,2)_{1/6}$, $u^c = (\bar{3},1)_{-2/3}$, $e^c = (1,1)_1$, $d^c = (\bar{3},1)_{1/3}$ and $L = (1,2)_{-1/2}$. All perturbative gauge and mixed gravitational anomalies vanish, the number of weak doublets is even, and the faithful image is

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}. \tag{12}$$

Among the 1,024 *subsets of the ten nontrivial irreducible exterior summands, nonemptiness, chirality and the four anomaly-trace conditions select exactly this fifteen-state mask and its charge conjugate.*

The exterior branching, anomaly traces and center kernel are finite calculations [11]. The supplied module is the familiar $\mathbf{10} \oplus \overline{\mathbf{5}}$ branching; this result does not require a physical SU(5) gauge field. It fixes neither three families nor their mass matrices.

8.2 The action and its coefficient realization

The configured ninety-mode fermion field intertwines all twelve source currents on full Fock space. Its Weyl matrices and sixty-state quantum links satisfy exact finite gauge identities. The sine regulator has eight valleys; its mirror multiplets differ from hole multiplets, and single-valley projection fails local gauge covariance. Three family copies, a Higgs doublet and canonical kinetics give tree coefficients $m_f = v\, s_f(Y_f)/\sqrt{2}$. Complete weak feedback converges to the specified mean family matrices in its pulse clock. A finite quantum Higgs extension preserves local Gauss identities with the same feedback. Vacuum selection, an interacting chiral limit and physical mass matching require separate control [12].

The retained pixel-dependent charged-lepton prescription gives coefficients $t_j(P) = m_j(P)/v(P)$. Three native port projectors P_2, P_3, P_{10} and a complete reset-memory process realize a positive matrix $M(P)$ satisfying

$$\operatorname{spec} M(P) = \{t_1(P), t_2(P), t_3(P)\}, \qquad M(P) = g Z_C.$$

Here Z_C is the unconditional C-duty coefficient, whose trace is the C-event probability p, and $g = \sum_j t_j/p$. Independent left and right probe durations give two admissible controller solutions, retaining disabled records. Summing every memory lag adds a coherent ordering potential and positive diffusion, and the twelve-address memory has an exact four-address recurrent encoding of all joint records [12].

8.3 Charged spectra of the source and the mass comparison

The open eight-site Weyl/contact cube at $c = 1/8$, $U = 1$ has a unique half-filled ground and 228 certified positive-weight lines per local one-spin addition or removal measure. A three-fermion probe reaches hidden spin 3/2, eta 1/2, outside linear response: 138 certified lines span 0.4024685–2.9747565 source units. Its 1,008-pulse instrument retains all 7,776 records and aborts at one ninth of the weight. On every connected open box all conserved even quadratics, including offsite and anomalous terms, are number and hidden spin, plus eta pairing exactly at centered chemical potential. This spin has no Lorentz assignment; a spread below eleven excludes the charged-lepton triplet under one unit [12].

A separate two-vertex gauge–Higgs control retains complete physical sectors of sizes 438 and 1596, with nonzero Higgs self-energy and exact Gauss identities. Six compensated probes have exact finite uncertainty and complete Ramsey records, including blocked creation and back-action. Their numerical frequencies are not physical masses: the graph, scalar cap, couplings, ground preparation and clock are supplied. Its finite record-sensitivity bound does not limit inference from known simulator specifications. Separate free overlap and local chiral-projector controls supply no common interacting chiral continuum [12].

The supplied pixel-dependent charged law has an exact C-duty memory realization. Inverse memory construction does not select its phase baseline, amplitude, scale or field census. Its forward coordinates give ratios 1 : 206.768300595 : 3477.365536596; the measured-fine-structure endpoint gives 1 : 206.768282739 : 3477.365262269. Only the latter lies inside both individual CODATA intervals, 206.7682827(46) and 3477.23(23) [21]. It inherits the measured coupling. The supersymmetric running closure gives 1 : 206.768297922 : 3477.365495531 under an extra field census;

the minimal census gives $1 : 206.782621310 : 3477.644439944$. These are conditional action ratios without controlled physical mass corrections [12]. The separate quark texture has $m_s/m_d \simeq 36.0$ and $m_u/m_d \simeq 1.005$, which no common unit repairs, so that texture is rejected.

The charged shape has an exact identity behind it. On each oriented face the regular C_3 action has the circulant commutant $C = aI + bR + \bar{b}R^2$.

Theorem 8.4 (Positive-chamber Koide identity)**.** *If $a > 0$, the eigenvalues $\lambda_k = a + 2|b|\cos(\theta + 2\pi k/3)$ are nonnegative and $\sqrt{m_k} \propto \lambda_k$, then*

$$Q = \frac{\sum_k m_k}{\left(\sum_k \sqrt{m_k}\right)^2} = \frac{1}{3} + \frac{2}{3}\Big(\frac{|b|}{a}\Big)^2,$$

so $Q = 2/3$ exactly when $|b|/a = 1/\sqrt{2}$, and the phase of b drops out.

Equal tracial event ranks and the specified square-root lift force finite balance. Applying it to canonical Yukawa masses requires $C \propto (Y_e^\dagger Y_e)^{1/4}$; all four current-gated memory duties fail this condition. A distinct coherent program with native exclusive weights $(10, 5, 5, 2, 50)/72$ produces $Y = C^2/3$. Its full-Fock vacuum compression retains every ancillary quantum outcome. The same coefficient has a controlled free Dirac limit with all lattice valleys retained; interacting chiral propagation and physical clock matching require separate proofs [12, 13].

A calibrated check sets $\theta \simeq 0.2222220471$ from the measured muon/electron ratio and gives tau/electron 3477.441636 without a tau input [19, 3]. The zero-momentum sea is phase-flat; the tested quantum-Higgs sector selects unit pass amplitude, zero phase and a repeated pair. For every attempt budget $n \geq 2$, reset C-duty gives $\lambda_2/\lambda_1 < 177$ and $\lambda_3/\lambda_1 < 784$, excluding direct lepton matching by that budget alone [12].

One contract fixes the Wilson source, Gauss polynomial state, reference and clock. Filtered probes and a retained origin give a relative observable algebra with commuting $H \geq 0$, P and complete Fourier readout. At eight cells, six paired sectors have positive thresholds but cannot directly yield the charged-lepton hierarchy. Six gauge multiplets have complete loop readouts. The rare-sector A_{1u} transition lies in $[63 - 59/1351350, 63 + 1/45675]$; its original weight vanishes under refinement. Weak-trion centroids have within-multiplet ratio below $53/48$. Dressed pairs at one-, two- and three-link separation have energy-window probabilities above $4/25$, $2/25$, $7/250$. Radial records resolve positive transfer above a fixed source cutoff, with capacity-dependent weight. All retain failures; none identifies a rest mass. Inversion pairs opposite native chirality with equal gauge and energy laws. Selected energy moments obey $M_2 \leq 4000000/a$, independently of link degree. A distinct apparatus matches the mean work with an elastic limit. Fixed-energy survival, relativistic thresholds and GeV masses remain unproved. A separate native-current construction gives one chiral family: seven massive Dirac color/lepton components and a neutral left-Weyl field in a specified Higgs background, with $m_c = |vy_c|$ and left-only weak currents. Its spectral regulator has a controlled free limit and finite nonlocality. A periodic quantum lepton/link/Higgs member has a unique ground on all 3424 neutral physical states and 2627 complete preparation/readout records. Conjugate charged thresholds coincide at 0.97009613 source units; an actual quantum Higgs composite has positive residue in the same line. Full-sector bounds give a local ground doublet; controlled weak hopping gives a unique eight-cell interacting ground, physical preparation and complete collective and charge-compensated fermion records. Their source energy bins require a propagation and mass identification [12]. A photon-containing target is globally gapless; pure Yang–Mills requires separate control.

The pixel fixed point uses the separate calibrated Standard Model census and response coefficients (Section 11.3).

8.4 Localized modes, topology and physical particles

A massive particle interpretation requires a localized excitation with persistent charges, a finite total energy, consistent exchange and decay rules, and a moving-state dispersion in the common clock. A topological invariant can constrain such states; it does not by itself fix their masses or identify their representation with an observed species.

For the radial two-spin source, an exact two-column kernel isometry W satisfies $hW = 0$. Extending the carrier by the native matter module and its three families, the local pairing $\Delta = C \otimes \overline{K}$, with $C = D_{\text{stagger}} i\sigma_2$, has an exact Nambu inclusion through which all twelve native currents intertwine. Choosing $Y_e = \sqrt{2}M(P)$ puts the retained lepton coefficients on these same localized states, and every complete record-driven pairing pulse preserves the inclusion. The bulk bands are additive $\lambda(h) \pm s_j(K)$, which supplies no relativistic mass shell. Three family copies are input, and creation of a bound fermion does not create the prescribed winding texture [12].

A compact alternative has two cores with shell degrees $+1$ and -1 and a constant exterior. Its spin energy and fermion vacuum-energy difference have bounds independent of exterior volume. For texture amplitude $M \geq 8$ there are exactly four localized modes, with explicit exponentially small outer-boundary errors. A complete rational certificate excludes zero from the four levels at the displayed separation-one source. Writing $A = W^{\dagger} h W$, the exact pairing inclusion retains $A \neq 0$; its bound energies shift the family coefficients. This compact band has four spatial columns, whereas the radial kernel has two. Neither its multiplicity nor its finite pair energy is an identification of particle generations or constituent rest masses [12].

A specified local texture instrument preserves both shell charges on every branch within an exact homotopy and spectral budget. Its supplied polar control transports the band, currents, pairing and interaction, with explicit control costs. Autonomous formation and moving-defect dispersion are unproved [12].

Classical unitary links preserve the compact 180-mode band and its quartic interaction. The separate sixty-state quantum link has exact Gauss-singlet tree transport, hopping $J\sqrt{d}$, linear string energy and measured default electric-to-hopping ratio one. Its uniform electric projector is not a gauge Casimir. The distinct Casimir-weighted gauge–Higgs action retains all 438 or 1,596 states. Complete color-loop records recover the default electric-to-magnetic ratio one. Exact elimination of the scalar complement gives a positive-resolvent action with additional transitions outside the original electric/magnetic span. The native raising link differs from a unitary Wilson coordinate. Physical strong coupling requires a common continuum action and renormalization map [11, 12].

Positive-field spin scaling supplies two canonical pairs and charged records of weight at least $1/774400$ in each narrow spatial-momentum cap, with broad energy support. The cubic-averaged centered ground at $U/c = 8$ has one even nodal measure shared by all eight transported valleys. Source locality makes a positive nodal atom equivalent to bounded-energy survival for packets within distance $O(a)$ of a node on one fixed normalized reference. Any full joint ballistic limit of those original readouts is dilation covariant and has no positive-weight nonzero mass shell. Identification of a massive shell therefore requires a justified source trajectory or another controlled reconstruction. Local sixth and eighth moments have exact source observables; their ground expectations are undetermined. A complete native dimer readout certifies spectral weight, without a particle assignment [12, 69, 9].

Mixing and Majorana vertices constrain additive winding charges, without identifying generations or physical spin-statistics. On the configured native 66-state source, two-record functions span every population direction; single records span only 42. Reflected moments of at most five attempts recover all 721 selective record maps and determine arbitrary finite record histories, with an explicit Gram-conditioning bound. This reconstruction uses the declared finite source and supplies

neither quantum tomography nor a physical particle-energy dictionary [12].

9 Interacting finite matter and controlled field limits

The reconstructed generator $-\log P$ has operator-norm distance exceeding 0.01984562 from every quadratic in its fixed occupation dictionary; this establishes finite interaction, without a relativistic limit. The executable specification also states instruments, memory and field services; its field actions and coefficients are inputs. Coherent ordering memory realizes the $\mathbb{Z}/2$ exchange-history charge of the native two-load configuration complex; its colored extension has fermionic exchange. Complete field factors then act on the full occupation algebra with a common calibrated clock and retained records. For projector-feedback memory, summing every stationary lag gives a coherent ordering potential and positive diffusion, with a fixed-time $O(h^2)$ unread-channel error bound for pulse step h. This semigroup requires stationary product preparation; it does not identify complete record tapes or physical masses [68, 12].

9.1 A nonquadratic dynamics on the localized algebra

On the compact 180-mode localized algebra of Section 8.4, choose the interaction $V = \lambda(N_W - 90)^2$, $\lambda \geq 0$, for each record Hamiltonian. It is parity even and commutes with every native internal current. Both CAR factors reduce each record Hamiltonian, exactly preserving the interacting localized algebra. For n pulses in time t, complete-memory contraction bounds the unread field and finite untouched reference's trace-norm error from the mean unitary by $\delta_n = O(1/n)$, for arbitrary initial memory correlations. The complete history error is at most $\min\{2, \sqrt{2\delta_n}\} = O(n^{-1/2})$. Adaptive instruments add interval bounds under the continuing memory law. The fixed algebra dimension makes these bounds uniform in exterior volume; the fixed texture supplies no translation-invariant many-defect field theory [12].

The interaction changes the spectrum. An exact sixteen-state reducing slice of the mean Hamiltonian has

$$H_4 = e_1(n_1 - n_2) + e_2(n_3 - n_4) + m(a_1^\dagger a_2^\dagger + a_2 a_1 + a_3^\dagger a_4^\dagger + a_4 a_3) + \lambda(N_4 - 2)^2,$$

and the two pair parities exhaust its levels. For the inherited three lepton coefficients and $\lambda = 1/128$, exact source enclosures select the odd/odd ground of each compact slice, where a positive-energy orbital has poles $e_1 + \lambda \pm m$, each with residue $1/2$. Removing e_1, e_2 changes the ground sector and its spectral law, and the paired even/even eigenstate has connected density cumulant $-\lambda/(4\sqrt{\lambda^2 + m^2})$. These slices do not determine the full interacting vacuum or laboratory particle masses [12].

A declared service prepares this Hamiltonian's faithful Gibbs reference, with $S' = \beta(E' - \langle X \rangle)$ and $F'' = -\beta C_\rho(X, X) \leq 0$ for a noncommuting coupling perturbation X. For $h_s = sT + MV$, differentiating the projected interaction gives $\lambda\{N_s - 90, d\Gamma(P_s' \otimes I_{45})\}$; its transported-band intrinsic force has norm at most 360, uniformly in exterior volume. These identities do not identify coupling or hopping forces with geometric stress, and Gibbs stationarity does not establish fixed-volume geometric entropy stationarity [9, 12].

9.2 What survives spatial refinement

At fixed spacing the Weyl/contact source has a positive-energy interacting ground and a time-reflection-positive Euclidean hierarchy. With $c = v/a$ and clock $H - UN/2$, the full eight-valley continuum is free for $0 \leq U = o(c^{2/7})$. In the opposite regime $U/c \to \infty$, original charged response leaves every fixed energy window. Holding the cell quartic coefficient $a^3 U$ at a positive constant

makes that weight $O(a^4)$. These sufficient bounds constrain possible interacting scalings; neither identifies massive Standard Model species [68, 12].

The scaling $c = J|E|^4$, $U = J|E|^8$ gives fixed-spacing interacting Heisenberg dynamics. Infrared bounds supply an ordered ground with $m_s > 1/1000$ and canonical fields with continuous translations, finite propagation and positive gapless energy. No sharp relativistic shell or surviving continuum interaction is proved. Their Gram constructs position conjugate to spatial momentum, with complete finite detectors and ordered source limits. Covariant detectors retain all time-evolved field responses without identifying particles; total-load coordinates are scalar on the protected spin sector. Clocks, spectral measurements and detectors retain their preparation and control premises [68].

At sufficiently small J/B, the declared spin source has an isolated lattice band with equal conjugate internal-charge energies. Its gap and locality bounds force zero physical speed under local refinement with bounded rest energy and J/B, excluding a massive relativistic limit on that clock and spatial map [12].

At $U = 0$, the ball modular charge of conserved Wick stress is $(2\pi/v)\int(R^2-r^2)\mathcal{E}/(2R)\,d^3x$. Nine independent shell preparations approach the same leading stress response with derivative control. Their relative entropy has an explicit sixth-order coefficient and an $o(R^4)$ nonlinear remainder in the stated lattice/small-ball regime. Responsive area and one common interacting state satisfying geometric stationarity are separate gravity premises [9].

10 The Einstein equation from a realized null balance

The gravitational hypothesis identifies Einstein dynamics with the coarse response of the same process.

Theorem 10.1 (Null tomography and metric ambiguity)**.** *Let V be four-dimensional with Lorentz form η of signature $(-,+,+,+)$, the opposite convention from Q in Section 7.2; nothing below depends on the choice. If a symmetric form X satisfies $X(k,k) = 0$ for every null vector k, then $X = \phi\eta$. Null-null values therefore determine a symmetric tensor modulo the metric line, and nine explicit null directions, the six axis rays $(1, \pm e_i)$ and $(1, s, s, \pm s)$, $(1, s, -s, s)$ with $s = \sqrt{3}/3$, reconstruct the quotient $\mathrm{Sym}^2(V^*)/\mathbb{R}\eta$ with design determinant $8192/27$ in the certificate's basis and a decoder obeying $\|x\|_\infty \le (2+\sqrt{3})\|q\|_\infty$.*

Proof. On future null rays $(1, n)$ the identity $X_{00} + 2X_{0i}n_i + X_{ij}n_in_j = 0$ on the unit sphere forces $X_{0i} = 0$, $X_{ij} = \phi\delta_{ij}$ and $X_{00} = -\phi$. The determinant and decoder bound are exact calculations, and the design, injectivity and ambiguity are formalized in Lean [1, 9]. □

Theorem 10.2 (Tensor completion of a realized null balance)**.** *Let (M, g) be a connected, time-oriented, C^3 Lorentzian four-manifold and T_{ab} a symmetric C^1 tensor with $\nabla^a T_{ab} = 0$. Suppose a common realized tower supplies $\kappa = 8\pi G_N > 0$ with*

$$G^{(g)}_{ab}k^ak^b = \kappa T_{ab}k^ak^b \tag{13}$$

for every null vector k at every point. Then there is a constant Λ with

$$G^{(g)}_{ab} + \Lambda g_{ab} = 8\pi G_N T_{ab}, \tag{14}$$

and a reference event at which the difference equals $-\Lambda_0 g_{ab}$ fixes $\Lambda = \Lambda_0$.

Proof. Theorem 10.1 applied pointwise to $G^{(g)} - \kappa T$ gives $G^{(g)}_{ab} - \kappa T_{ab} = \phi g_{ab}$. The contracted Bianchi identity, the Ward premise and metric compatibility give $\nabla_b \phi = 0$, connectedness makes ϕ constant, and $\Lambda = -\phi$. □

Null balance fixes Einstein's equation up to a metric term; the Bianchi identity and stress conservation make its coefficient the cosmological constant. The coefficients of entanglement-equilibrium arguments [61] enter through one machine-checked identity: scalars with $\delta S_{\mathrm{bulk}} = \frac{8\pi^2\ell^4}{15} t$, $\delta A = -\frac{4\pi\ell^4}{15} f$ and $\delta S_{\mathrm{bulk}} + \delta A/(4G_N) = 0$ obey $f = 8\pi G_N t$. For OPH to supply (13), one common refinement tower must realize (G1) the effective manifold of Proposition 7.6; (G2) geometrically normalized cap modular flow whose directional charges define one stress tensor; (G3) its Ward identity, independently of that reconstruction; (G4) the edge-center entropy split, the normalization $\delta\langle Z_L\rangle = \delta A/(4G_N)$, the modular first law and generalized-entropy stationarity; (G5) a small-ball family with controlled remainders and enough observer directions; and (G6) universal coupling, a vacuum reference and a scale calibration fixing $\kappa = 8\pi G_N$. The finite layer contains null tomography, the edge-entropy split and the coefficient identity, and the machine-checked composition proves the tensor step from an explicit null-balance premise; the bridge between them is (M4).

Section 9 supplies sector-relative modular and Ward inputs to (G2)–(G3) for the proved free Weyl sector, and the matter first-law and small-ball parts of (G4)–(G5). Nine explicit shell perturbations span traceless stress variations; finite boosted-ball responses have full rank for $0 < R/L \le 2^{-28}$. At exactly $U = 0$, compact lattice states and quadratic detectors recover the leading stress response in a joint spacing/small-ball limit. On each fixed ball, the actual finite-density modular flows also converge on all fixed probes, with a generator graph core and convergent selected relative entropies. The proofs retain all eight valleys and do not assume a uniform covariance gap [69]. The Wilson source supplies nonzero energy-current response to a finite pulse, with complete records and positive work. Its energy Ward identity retains applied work. The regional modular first law holds on supported marginals. A physical comparison bounds correlations with the polynomial-state remainder. Geometric stress, uniform area response and universal coupling with calibrated G_N require source maps [9].

11 Cosmology and the closure coordinates

The cosmological hypothesis asks the same process to fix the vacuum, the dark sector and the primordial response. In this paper the layer consists of conditional branches and diagnostics of declared maps.

11.1 The dark sector: non-luminous charge, the deep galaxy law, and the vacuum coordinate

Under supplied universal-coupling, physical-stress and source-localization premises, sourcing gravity by modular charge classifies any geometric excess over the luminous term as non-luminous modular charge, with anomalous modular energy on overlap collars as one candidate carrier [15]. Conditional on two named premises the deep-regime profile is characterized rather than supplied: an enclosed-mass law that is degree-one scale covariant and composes independent sources in quadrature carries a unique constant $a_0 > 0$ with $M_A(r) = r\sqrt{M_b a_0/G}$, which forces flat rotation curves and gives $v^4 = GM_b a_0$, the baryonic Tully–Fisher relation, each premise load-bearing by an explicit counterexample. A per-cut model gives $a_0 = Gn^2 c$, with only the product $n^2 c$ identifiable, and the value of a_0 is fixed by no source law. Reading the dark-energy density as $\rho_{\mathrm{DE}} = \kappa/N$ at constant κ gives $w = -1 + \frac{1}{3}\,\mathrm{d}\ln N/\mathrm{d}\ln a$, so fixed capacity gives the exact point $(w_0, w_a) = (-1, 0)$ and

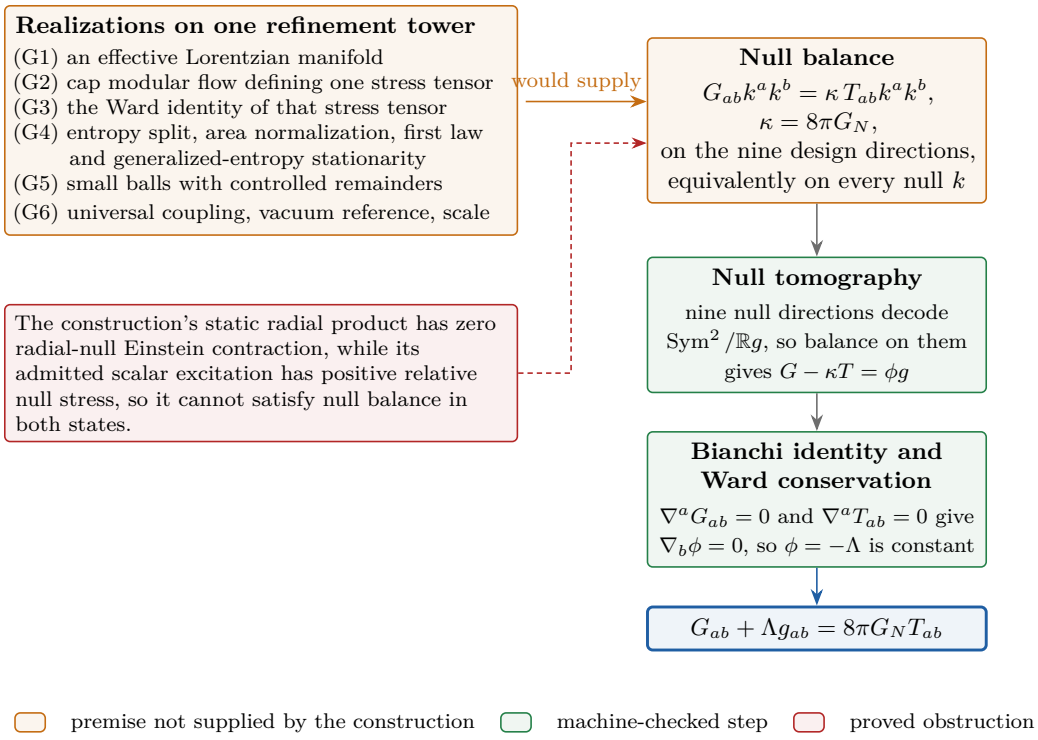


Figure 3: The Einstein equation as tensor completion of a realized null balance (Theorem 10.2). The tensor step is machine-checked from an explicit null-balance premise. Realizations (G1) to (G6) on one refinement tower would supply that premise, and no complete realization of them on one common refinement tower is established; the tested static radial product fails the balance for its own scalar excitation.

nondecreasing capacity gives $w \geq -1$; exclusion of that point fails the fixed branch, while agreement supports nothing, since it is also the ΛCDM null [16].

11.2 Expansion, horizons and records

Expansion adds no causal data to the layered read law. A physical read $\|\sigma_{j+1}t - \sigma_{j+1}s\| \leq \sigma_{j+1}a$ holds exactly when $\|t - s\| \leq a$, so every scale-factor profile generates one order, as conformal invariance of causal structure requires [42, 43]. The profile enters only the count measure: a constant physical event density gives the four-volume weight $\sigma^4\, d\eta\, d^3x$, and the source law does not select σ [17]. On a finite screen of fixed capacity, maximizing generalized entropy over sector probabilities gives $S_{\mathrm{gen}}^{\max} = \log M$, and uniform transfer of a fraction f of the horizon capacity to an observer changes that maximum by $\log(1 - f) < 0$, the sign of the de Sitter time advance [14]. Public records keep an exact information ledger: pinching is an average of independent sign unitaries whose entropy gain equals the relative entropy to the pinched state, and on a supplied three-dimensional layered population crossing reads obey an area law with coefficient $(\pi/4)n^2a^4$. A declared nat per crossing calibrates that area; horizon entropy, Hawking flux and evaporation need the native channel and a physical geometry [18].

11.3 Pixel closure and the common particle response

The dimensionless pixel coordinate is $P = a_{\mathrm{cell}}/\ell_P^2$. Its self-reference equation relates it to a response computed from the chosen source specification,

$$P = \varphi + \frac{\sqrt{\pi}}{A(P)}, \qquad \varphi = \frac{1+\sqrt{5}}{2},$$

where A is the declared inverse electromagnetic-coupling readout. A fixed-point proof determines a root of that map. Identifying its readout with the physical Thomson coupling requires the current response of the same action, vacuum, spectrum and clock.

One explicit conditional map joins the native matter census to the charged-lepton law of Section 8. For unified coupling u, energy unit E_* and scale coefficient β, take

$$\frac{M_U}{E_*} = e^{-2\pi} P^{1/6}, \qquad \frac{v}{E_*} = P^{-1/2} e^{-2\pi/(\beta u)}, \qquad \alpha_i^{-1} = u^{-1} + \frac{b_i}{2\pi} \log \frac{M_U}{m_Z}.$$

The infinite heat-sum equation, with certified tail bounds, supplies u; the same neutral-Higgs relation supplies m_Z/v. Exact charge and Casimir traces on the fifteen native matter rows, with three families and one complex Higgs doublet, give

$$(b_1, b_2, b_3) = (41/10, -19/6, -7)$$

when inserted into the declared four-dimensional one-loop functional. The representation traces are finite consequences of the gauge module; the loop functional and scale law are additional physical modeling inputs. This is the intended surviving chiral census; applying it to the eight-valley lattice requires a proved mode-removal or threshold map [11, 10, 12].

For every charged lepton and all six quarks, the same coefficient masses enter the response kernel

$$K_f = \frac{2N_c Q_f^2}{\pi} \int_0^1 x(1-x) \log\left(1 + \frac{m_Z^2}{m_f^2} x(1-x)\right) dx,$$

$$A = \alpha_2^{-1} + \frac{5}{3}\alpha_1^{-1} + \sum_\ell K_\ell + \left(1 - \frac{3\alpha_3}{\pi}\right) \sum_q K_q.$$

The quark multiplier is a prescribed screening factor. This map includes neither complete charged-vector and scalar response nor a nonperturbative hadronic spectral measure, so it is not the full physical electromagnetic running of the Standard Model.

The minimal census with $\beta = 4$, positive couplings, $P \geq 1$, $M_U \geq m_Z$ and no intermediate thresholds has no solution: the scale equations force $\alpha_2, \alpha_3 > 1/6$, whose combined heat sum is below $1/80$, while pixel closure requires at least $1/4$. With the explicit alternative $\beta = 7$, interval bounds certify a locally unique self-consistent root

$$P \simeq 1.628533369611833, \qquad A \simeq 168.815082928453.$$

Infinite heat tails, implicit-root derivatives and a strict interval self-map are controlled. The memory inverse is certified over the same root enclosure, so its two admissible controllers realize those same charged-lepton coefficients. This is a consistent conditional coefficient map; its inverse coupling is not an identification with the measured fine-structure constant [12].

A complete physical loop must replace the coefficient readout by the current spectral response of the actual interacting states. Compact binding energies and quartic interactions change that spectrum. Therefore closing the displayed scalar map does not close the joined physical particle-response

loop. Detector use of $g(P) = \sqrt{4\pi\alpha(P)}$ reads a supplied coupling; it does not independently derive its electromagnetic meaning.

A separate exact scalar is due to Wissner-Gross [20]. For four slots of dimensions $(3, 2, 3, 2)$, traceless letters on unordered slot pairs give 169 product records and three exchange-odd directions complete a 172-dimensional feature space. Two commuting signed exchanges X, Y define $K = (2I - X)(3I - Y)/2$, whose joint projectors have gains $1, 2, 3, 6$ and ranks $79, 48, 42, 3$. With the declared normalization 31104,

$$\alpha_{\text{pair}} = \frac{79 + 48\sqrt{2} + 42\sqrt{3} + 3\sqrt{6}}{31104}, \qquad \alpha_{\text{pair}}^{-1} = 137.035999189566\ldots,$$

against the measured 137.035999177(21) [21]. An independent reconstruction verifies the arithmetic with no continuous coefficient. Its identification with the Thomson coupling, its normalized-trace readout and its attachment to the screen grain are conditional [10].

The other closure coordinate, record capacity N, bounds the declared correctable record algebra. Under its de Sitter entropy interpretation,

$$N = \frac{3\pi}{\Lambda \ell_P^2}, \qquad \ell_P^2 = \frac{\hbar G}{c^3}. \tag{15}$$

This dictionary converts a capacity into a cosmological constant after physical scale calibration. It does not make a fitted capacity an independent cosmological prediction. The specialist analyses separate conditional root certificates, measured-coordinate comparisons and independent physical tests [10, 16].

12 Realized results and physical obligations

Physical claims require one realization carrying compatible events, fields, action, states and readouts across refinement. The following results state their domains, premises and exclusion criteria. The target is equality of limiting observable laws; finite tests retain approximation and measurement errors.

12.1 Realized finite domains and physical extension

Table 2 pairs each constructed answer with its physical conditions. Existence and subjective experience are interpretive targets: public reports of experience have operational meaning, but do not derive the character of experience. The common specification gives these diverse questions a single testable setting. Its physical correctness depends on satisfying the conditions in the right column together, including the measured particle spectrum and responsive geometry, and on surviving independent prospective tests.

Table 2: Explanatory scope and status of the proposed common specification.

Question	**Constructed result**	**Physical or interpretive condition**
Why anything exists	An inhabited self-recovering history is the closure hypothesis; an exact finite consensus closure has fixed packets	A universe-level fixed point and its identification with our history are not established by the finite fixed-point theorem
Why there are public facts	Protected overlap records have canonical normal forms under the displayed repair hypotheses	One source must realize the observer, scheduler and comparison contracts across the claimed domain
Why experience occurs	The executable observer has defined access, memory, intervention and observer-relative record interfaces	These interfaces give no derivation of qualia or of why any record has phenomenal character
Why space has three directions	The twelve-port response has an exact three-dimensional slow band and Euclidean completion	Physical position, length and a cofinal spatial refinement require the realization map (M1)
Why observations are quantum	Completed records obey Born, conditioning and correlation identities in a finite complex algebra-state model; native histories reconstruct a positive functional on the full matrix algebra	The source must produce that algebra, its tomographic effects and its physical preparations; the reconstructed functional is a history ensemble and classical consensus alone does not imply them
Why entropy increases	Fair-coin seam repair is exact conditional resampling with entropy production, complete-tape accounting and a native record-cost instance of the finite four-law core	Laboratory heat, temperature, physical energy exchange and a strain law need their own attachment
Why these forces and particles	Complete-response premises force $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$; a chosen matter module has an anomaly-free generation; native currents, modes and mass coefficients share a finite operator construction, and the interacting cube has certified charged spectra	Physical chirality, dynamical gauge fields, stable defect identities and the complete measured mass and mixing spectrum require a common interacting limit
Why fields propagate in $3+1$ dimensions	Controlled free fermion and conditional scalar limits, together with a finite interacting core, have explicit domains	Their states, clocks and interactions need one controlled relativistic continuum limit
Why geometry responds to matter	Null tomography and tensor completion yield Einstein dynamics from the stated common null-balance and entropy premises	The same source must realize the geometry, conserved stress, clock and generalized-entropy response
Why cosmological observables take their values	Finite capacity, response and data comparisons supply conditional relations and tests	A selected physical vacuum, calibrated scales and independent prospective comparisons are required

The physical extension requires autonomous localized excitations, a controlled chiral spatial limit, dynamical gauge fields and a common geometric stress and entropy response. The table summarizes those requirements without treating a finite certificate as empirical identification.

12.2 Results and obligations by layer

Table 3 gives the result and obligation of each layer. "Established" means proved on its stated domain or executed with retained receipts; "conditional" means that the conclusion depends on displayed antecedents beyond its realized finite instances; the open join is what the executable model must supply on the same process; and the stopping test is the result that would remove the layer, its branch or the tested law. Negative results are findings about the tested law, stated with their exact scope.

Table 3: Established constructions, physical obligations and negative controls.

Layer	**Established and conditional**	**Physical obligation**	**Negative results and stopping test**
0. Closure and agreement	Established: on the finite consensus branch the closure map is an affine idempotent fixing exactly the packets on consensus normal forms; the record-closure equation is posed exactly. Interpretation and acceptance criterion, not a theorem	A history domain and closure operator for the executable model; interpretation maps fixed independently of validation outcomes, compatible across observers and refinement	The bounded completion class does not identify the solution N; no theorem selects a unique universe. A contradiction among indispensable premises stops the proposal
1. Repair mathematics	Established: canonical observable normalizer; consensus normal form under the transactional diamond; adaptive-run, capacity and fixed-federation bounds; Born weights without continuity axiom, Lüders selection, Tsirelson bound, no-copying; entropy laws of fair-coin repair; positive residual mismatch from fixed holonomy. Conditional: the finite algebra-state representation of completed records is an input	Native justification of the complex effect representation, equal-effect operational equivalence and disjoint coexistent refinement; an implemented scheduler within the specified process	Scheduler choices remain explicit inputs; deciding consistent extension is NP-complete for succinct Boolean systems; retained routing logs do not identify federation gluing. As theorems they are refuted only by a counterexample
2. Carrier and repair law	Established: the seam rule from four requirements; the classified convex record-law family; the gap $(5-\sqrt{5})/60$ for every admissible table; the rank-three slow-response completion; one seam per native repair event. Conditional: physical position consumes (M1)	One specified record law and federation gluing validated against the physical process; a physical length	At full-port energy scaling the refining screen loses the rank-three band. The C/E instrument test separates C, E and the current-only law; a resolved dispersion outside the band excludes the whole carrier class

Table 3 continued

Layer	Established and conditional	Physical obligation	Negative results and stopping test
3. Executable observer	Established: native observer with record-caused feedback; electromagnetic and scalar detectors; one interacting process with certified infinite-occupation contrast; overlapping observers exact on 5,768 output words; local event clocks; identification of the record law; coherent observers on matched screens; complete $O(1/d)$ error for the smooth four-read protocol; a continuing C/E finite-buffer observer with counted exports and an exact stationary source/complete-record marginal	One continuing autonomous observer with every service, memory, reset and resource accounted; the declared apparatus, schedules and preparations implemented on that same process	Outcome accuracy does not certify the output state; committed native records form a classical process, so correlations beyond $S = 2$ need declared coherent preparation services
4. Fields and clocks	Native screen operator with spectral dimension two; conditional smooth static $(1 + 3)$ scalar readouts; centered eight-valley free fermion limit; ordered gapless spin fluctuation limit	Compatible physical clocks, state and energy normalization; an interacting chiral $3 + 1$ field with the observed massive and massless sectors	Passive repair has no wave sector; screen spectral growth excludes a direct elliptic three-manifold identification; a gapless spin field is not a proved relativistic interacting particle theory
5. Gauge and matter	Conditional Lie-type forcing and finite matter image; common native currents, localized modes and pixel-dependent mass matrix; compact opposite-degree pair; exact interacting finite response; certified 228 linear and 138 composite charged lines; complete conserved quadratics	Autonomous finite-energy particles with the observed charges, spin, statistics, mixing and pole masses on the same action and clock	All twelve mass comparison values lie outside their printed intervals; the six-power quark texture is rejected. Winding does not pin zero energy; compact binding and quartic interactions shift the spectrum. The scalar pixel root does not close the physical interacting current-response loop
6. Gravity	Established: null tomography and the tensor completion from an explicit null-balance premise; physical energy continuity, boundary-completed spatial strain and polynomial-state modular work bounds. Conditional: the Einstein equation up to Λ under (G1) to (G6)	(G1) to (G6) realized on one refinement tower with the same matter, modular flow and clocks; defect maintenance energy in the same stress	The static radial product fails null balance for its own scalar excitation; stationary repair entropy is distinct from fixed-volume generalized entropy. Failure of (M4) removes the spacetime and Einstein conclusions together

Table 3 continued

Layer	Established and conditional	Physical obligation	Negative results and stopping test
7. Cosmology and tests	Established: the deep-law uniqueness under two premises; $(w_0, w_a) = (-1, 0)$ at fixed capacity; the de Sitter sign; the dispersion band. Conditional capacity and pixel fixed points	A common physical response for P, capacity and scale calibration, and independently fixed observation maps	Minimal matter with the specified scale coefficient four has no pixel root; coefficient seven has a root whose response is not the measured coupling. Exclusion of the band, a C/E rejection, a charged-lepton average beyond three standard uncertainties from the balanced center, or exclusion of $(-1, 0)$ fails its branch

12.3 The realization maps

Five maps name what a physical realization must supply, and each boundary statement in this paper cites them by label.

(M1) Position and scale. An identification of the intrinsic carrier completion with physical position, with a physical length, cofinal overlap gluing between carriers, and a refinement limit (Section 5).

(M2) Clock and energy calibration. A common comparison between internal evolution, laboratory duration and energy in measured units. The source-executed reference clock measures the same controlled intervals as the field, and the corresponding energy observables have complete detectors. Extending and validating that calibration across the physical process remains necessary. Native event counts are a different clock; the rejected direct count-to-optical-phase attachment in Section 13.3 does not reject the calibrated reference clock.

(M3) Laboratory current. An identification of the finite port response with a physical gauge current, with a physical matter action and a continuum operator limit.

(M4) One common physical tower. A single physical refinement family realizing the event, modular, stress, entropy, continuum, vacuum, and scale premises together, rather than several independently declared islands.

(M5) An implemented and matched action. A specified field and interaction law realized by the same native services, with its complete state space, relative normalizations and operational readouts. Parameters may be declared or calibrated; which predictions follow without consuming their measured targets must be stated. Physical identification requires compatible propagation, currents and interactions across refinement.

The finite constructions realize restricted instances of these maps: a calibrated common clock, internal current operators, a full field action, energy readouts and compatible reference observations. A common physical extension and validation, with the requirements of Table 3, is not established by those restricted maps. A theorem that consumes none of them stands on its own finite domain. Section 13.3 tests two simple attachments of native records to published data. The construction paper gives the detailed premise map, including the retained negative controls that bound how the carrier, the response triplets, the feedback class and the kinetic term may be attached [68].

13 Empirical meaning and falsifiability

The custody-bound register fixes target definitions, exclusion thresholds, precision floors, exposure classes and decision rules before eligible comparison [3]. Two kinds of test follow from the construction: a propagation anisotropy fixed by the carrier's symmetry, and an instrument comparison fixed by the executable observer. Published records also test precise attachments of native records retrospectively.

13.1 Propagation anisotropy at angular rank six

Angular rank means spherical-harmonic rank j, not a particle spin. The dimension of the invariant subspace of the sixty proper icosahedral rotations in angular rank j is

$$m_j = \frac{1}{60}\Big[(2j+1) + 15\,U_{2j}(0) + 20\,U_{2j}(\tfrac{1}{2}) + 12\,U_{2j}(\tfrac{\varphi}{2}) + 12\,U_{2j}(\tfrac{\varphi-1}{2})\Big],$$

with U the Chebyshev polynomials of the second kind, and the exact table gives $m_1 = \cdots = m_5 = 0$ and $m_6 = 1$ [56].

Theorem 13.1 (Icosahedral angular-rank universality)**.** *Let the carrier kinetics be any operator $\lambda_a(k) = a^{-2}\sum_d w(d)\,[1-\cos(a\,k\cdot d)]$ with a finite direction multiset and weights invariant under the proper icosahedral rotation group. Then the dispersion is exactly isotropic through angular rank five at every order in a, every directional term below angular rank ten is one multiple of the unique normalized invariant $\mathcal{I}_6$, and whenever the weighted sixth-moment coefficient is nonzero the first directional artifact appears at order a^4k^6, suppressed by exactly $1/16$ under one binary refinement at that order. Rank six is the least symmetry-allowed nonzero anisotropic rank in the class.*

Proof. The order-$a^{2m-2}k^{2m}$ angular content is an invariant polynomial of degree $2m$ with harmonic components at even ranks $j \le 2m$, and the invariant table forces zero at $L = 2, 4, 8$ and a one-dimensional space at $L = 6$. The table, the 62-direction stationary census, the orbit verification and the nonvanishing sixth-moment coefficients are certified in exact arithmetic [1]. □

Proposition 13.2 (Primitive twelve-port branch prediction)**.** *Suppose the intrinsic spatial kinetic symbol lies in the cosine class of Theorem 13.1 with scale-independent coefficients, the complete orbit of the twelve port directions u_i is its sole hop support through order k^6, the carrier scale a is finite and positive with the quadratic term normalized to k^2, one carrier rest frame is transported coherently over the experiment, and the tested scalar or polarization-independent sector realizes the operator. Then*

$$\omega^2 = \frac{1}{2a^2}\sum_{i=1}^{12}[1-\cos(ak\,u_i\cdot\widehat{k})] = k^2 - \frac{a^2}{20}k^4 + \frac{a^4}{840}k^6 + \frac{2a^4}{7875}k^6\mathcal{I}_6(\widehat{k}) + O(a^6k^8), \tag{16}$$

with $\mathcal{I}_6(n) = \frac{25}{132}\sum_i P_6(u_i\cdot n)$, so in a transported frame the branch predicts the scale-free relations

$$\frac{B_6}{C_4^2} = \frac{32}{315}, \qquad \frac{B_0}{C_4^2} = \frac{10}{21}, \qquad \frac{B_6}{B_0} = \frac{16}{75}, \tag{17}$$

and every intrinsic anisotropic coefficient at $1 \le j \le 5$ vanishes.

Proof. The exact carrier moments $\sum_i(u_i\cdot n)^2 = 4$, $\sum_i(u_i\cdot n)^4 = 12/5$ and $\sum_i(u_i\cdot n)^6 = 12/7 + (64/175)\,\mathcal{I}_6(n)$ substituted into the cosine series give the expansion; eliminating a gives the ratios. □

A second frozen ray follows from the source seams.

Proposition 13.3 (Source-seam edge ray and conditional propagation branch)**.** *The boundary current of the complete thirty-seam source packet maps through the antipodal-odd load readout onto the even-sum module D_6 of Theorem 5.1, with Smith invariants* $(1,1,1,1,1,2)$*, and every directed seam defines one translation of the internal record carrier whose complete normalized direction multiset is the thirty-direction edge orbit. If the feasible move laws and their objective are natural under the proper carrier action and A3 supplies a unique normalized minimizer, transitivity forces weight* $1/60$ *on every directed seam, and the selected internal operator is the source-counting homogeneous convolution.*

Suppose these record translations are physical displacements under one homogeneous position action, that the complete edge orbit is the sole direct support through the displayed order, that proper-carrier covariance fixes one common weight, that the same operator acts on one scalar or polarization-independent physical sector with quadratic term normalized to k^2, and that the action carries the stated gluing, scale, frame, readout, nuisance, and exclusivity data. Its spatial kinetic symbol is then

$$\Lambda_a(k,\widehat{k}) = \frac{1}{5a^2}\sum_{j=1}^{30}\Big[1-\cos\Big(ak\,w_j\cdot\widehat{k}\Big)\Big], \tag{18}$$

with $C_4 = -a^2/20$, $B_0 = a^4/840$ and $B_6 = -a^4/12600$, hence $B_0/C_4^2 = 10/21$, $B_6/C_4^2 = -2/63$ and $B_6/B_0 = -1/15$. The intrinsic anisotropic coefficients at angular ranks one through five vanish, and the rank-six vector lies on the rotated $\mathcal{I}_6$ orbit.

Proof. Exact integer incidence gives the displayed image and Smith invariants, and the response-selected Gram pullback gives the completion of Theorem 5.1. The sixty directed labels map two-to-one onto the thirty signed edge directions, whose unit-normalized moments are $\sum_j (w_j\cdot n)^2 = 10$, $\sum_j (w_j\cdot n)^4 = 6$ and $\sum_j (w_j\cdot n)^6 = 30/7 - (2/7)\,\mathcal{I}_6(n)$; substitution in (18) gives the coefficients and ratios. The incidence, quotient, orbit binding and moment identities are machine-checked in exact arithmetic [1]. □

The proposition contains an exact finite source theorem and a conditional physical branch. The naturality and minimizer clause supplies a homogeneous internal action whose generator satisfies the positive maximum principle and an exact Dirichlet identity, and whose plane-wave spectrum is the symbol above. The edge branch and the primitive-port branch have opposite rank-six signs, and their coefficient rays, custody packages, exposure exclusions and comparison budgets are separate.

Theorem 13.4 (Carrier-class dispersion band)**.** *For the declared positive-weight scalar cosine class over orbits O_s of the sixty proper carrier rotations with radii $r_s > 0$, weights $w_s > 0$ and the k^2 coefficient normalized to one, write $\Lambda_{\mathrm{norm}} = k^2 + C_4k^4 + (B_0 + B_6\mathcal{I}_6(\widehat{k}))k^6 + O(k^8)$ and $\mu_m = \sum_s w_s|O_s|r_s^m$. Then $C_4 = -(a^2/20)\mu_4/\mu_2 < 0$ for every member; $B_0/C_4^2 = (10/21)\mu_2\mu_6/\mu_4^2 \geq 10/21$ with equality exactly on single-radius members; the anisotropic ranks one through five vanish; and B_6/B_0 lies in $[-16/135, 16/75]$, the face and vertex orbits attaining the endpoints and the edge orbit sitting at $-1/15$.*

Proof. Summing over the sixty rotations gives $\sum_g ((gu)\cdot\widehat{k})^6 = \frac{60}{7} + \frac{64}{35}\mathcal{I}_6(\widehat{u})\mathcal{I}_6(\widehat{k})$ on unit vectors. The floor is the Lagrange identity $\mu_2\mu_6 - \mu_4^2 = \sum_{i<j} W_iW_jr_i^2r_j^2(r_i^2 - r_j^2)^2 \geq 0$ with $W_s = w_s|O_s|$, and the band holds because the rank-six numerator is a positive-weighted mean of seed values of $\mathcal{I}_6$ in $[-5/9, 1]$ [1]. □

A resolved intrinsic dispersion with B_0/C_4^2 below $10/21$, or B_6/B_0 outside $[-16/135, 16/75]$, or a residue at ranks one through five, excludes every member of the declared class at once. The primitive-port relation is a frozen prospective conditional physical-branch prediction: its target statement, coefficient snapshot, registration record, public commit custody and detached calendar attestations fix the prospective content, the dated template search whose icosahedral templates returned a family-wide p-value of 0.64 is ineligible, and a verdict requires a dataset-specific registration with a joint likelihood, frames, orientation profile, nuisance model and calibrated threshold. The branch fails at five standard deviations or more if an isolated intrinsic C_4 is positive, a lower-rank coefficient is nonzero, or the calibrated profile likelihood excludes the complete manifold $C_4 < 0$, $B_0 = \frac{10}{21}C_4^2$, $B_6 = \frac{32}{315}C_4^2$. A null is inconclusive, since the branch supplies no positive lower bound on a, and a failure rejects the primitive twelve-port propagation branch; it reaches the framework as a whole only after a stronger source law makes that branch forced and exclusive [3].

13.2 An instrument test from the executable observer

Under a common current-record interface every admissible record law has the classical statistics of C and E. The multiplier g, the coherence between an empty seam and a waiting single load, separates C, E and the law that records only the current. On a fixed coherent two-state source preparation the readout probability is $3/8 + \frac{1}{2}\,\mathrm{Re}(g\,e^{-i\theta})$: reversing the phase moves C from $5/8$ to $1/8$, E predicts $3/8$ for both, and the law that records only the current predicts $3/8 \pm \sqrt{2}/4$ [70]. Four fixed settings include common-phase and incoherent controls. The complete-count evaluator uses 8192 trials per setting, an independently established probability-error budget of $1/64$ and an exact sampling certificate below $1/1000$; it rejects malformed or incomplete records and reports *no data* when no independent counts are supplied. A source-executed implementation runs both complete 78-state instruments, the four preparations and a complete Lüders readout from twelve-port control words, retaining every record and attempted trial [68]. A sequential design on one continuing carrier needs neither exact resets nor independent trials: 379 discarded native attempts before each of 3,927 trials replace the reset, failed heralds count as trials, and conditional-law bounds on the dependent transcript give a wrong-candidate probability of at most $3/400$ under the exact-source model [89]. Synthetic controls exercise the decision rules and establish no physical verdict; physical encoding, independent calibration and comparison data are the open requirements.

13.3 Native records against public data

Published records can test precise attachments of native records without a new laboratory realization; each test below is retrospective, states its readout hypotheses, and assigns no prospective verdict.

The diagonal native branch is a classical stochastic process, so under independently chosen settings and local factorized responses it obeys the CHSH bound $S \le 2$. The deposited records of the loophole-free Hensen experiment [92] reproduce 245 eligible trials, 196 wins and $S = 2.42250$, and the published fixed-count analysis gives a tail bound near 0.03917 with its setting-calibration allowance [90]. This rejects the local classical reading of the diagonal branch under that experimental contract. Committed native records form a classical process: the native grammar keeps diagonal inputs diagonal under every classical feedback history, and two implementations with the same committed branches, which give $3/4$ and $1/4$ under a read before commitment, give $1/2$ for both once the record is committed [98]. Correlations beyond $S = 2$ therefore need uncommitted coherence from declared preparation services, and every admissible record law attenuates it [70]. The replay leaves the completed record surface untouched, on which a declared entangled state

reaches the Tsirelson bound of Theorem 4.5, and it does not distinguish C from E.

Event clocks need no external validation, and identifying their readings with a published measurement is a testable hypothesis. Attaching the published 300-second blocks of the Ar^{13+}/Yb^{+} optical frequency ratios of King et al. [93] to the normalized incident-repair count ratio of two native ports or regions, with a declared gain and frequency bracket, gives a complete-count probability below 10^{-104} for two ports and 10^{-22} for every pair of the 3,433 regional event sets [91]. The direct attachment is rejected together with its readout assumptions; the native event clocks are unaffected as clocks of the model. Published trapped-ion records separately test a supplied oscillator model executed through the source interface [69].

13.4 Diagnostics and what fails together

The other quantitative comparisons carry their own decision shapes and do not count as predictions. The closure roots of Section 11.3 are diagnostics of declared maps. The registered conditional primordial branch $n_s = 1 - P/48 = 0.96602$, at the calibrated grain P, lies at the lower 0.612 percent position of the official ACT DR6, Planck and DESI joint posterior chains, whose mean is 0.9743 ± 0.0034 [94, 91]. Those data precede the registration, so this is a retrospective tension under baseline ΛCDM, not a likelihood ratio, and the registered comparison with a later dataset is unchanged. A charged-lepton world average more than three standard uncertainties from the balanced-circulant center refutes the equal-rank-two tracial reading of the face-circulant sector; a resolved deep-regime galaxy sample violating degree-one radial scale covariance or quadrature composition removes the uniqueness of the deep law; and exclusion of $(w_0, w_a) = (-1, 0)$ fails the fixed-capacity branch. The conditional implications share premises, so they fail in groups: failure of (M4) removes the effective-spacetime and Einstein conclusions together, failure of (M3) removes the gauge and matter interpretations together, and failure of (M2) removes every statement in laboratory units. The finite normal-form, invariant-theory and algebraic results retain their stated domains under all three.

14 Machine verification and reproducibility

The results have different logical types, and Table 4 gives the classes used throughout; each result throughout the paper carries one of them. Each step is identified by its artifact as a Lean theorem, exact executable certificate, conditional paper argument, numerical receipt, or explicit unproved premise.

Each result maps to an artifact class: a Lean library with no admitted propositions in the dependency closure reported here, including formal premise boundaries and countermodels; exact-arithmetic code receipts, integer and rational computations and outward-rounded interval certificates, each emitting a canonical hashed payload; executable observer receipts that replay every retained result and check its artifact hashes; simulation receipt bundles pinning the source revision, configuration, seed where stochastic, tolerances and output hashes; verifiers that recompute verdicts from clause vectors and reject receipts whose stored verdict disagrees; and adversarial negative controls, so that every certificate ships with mutations that must be detected. Machine-readable records classify every quantitative row as a source result, reconstruction implication, diagnostic, prospectively fixed test or rejected candidate, and reject any stronger classification unsupported by the recorded dependencies.

Table 5 names the artifact family behind each group of results. The repository, receipts, schemas and rebuild instructions are public [1, 2, 69]. The formal and executable stack checks the claims

Table 4: Epistemic claim classes used in the manuscript.

Finite theorem	A deductive result proved from displayed premises on a stated finite domain. An exhaustive computation counts as part of the proof only with a specified domain and a soundness argument.
Certified finite computation	An executable witness whose inputs, controls, and output are fixed by a reproducible certificate; execution alone does not promote its interpretation to a theorem.
Conditional analytic theorem	A theorem whose hypotheses include a supplied continuum action, state space or limit; its conclusion holds on that supplied object.
Source limit theorem	A theorem about the limit of a declared family of finite source executions along a stated subsequence, with its normalizations and order of limits displayed; the limit is derived, and its physical identification is a separate claim.
Reconstruction implication	An exact implication whose antecedents include displayed geometric or physical premises.
Unsupplied realization map	A required map from a finite object to a physical event, current, field, clock, scale, or continuum structure for which no construction is asserted.
Diagnostic	A certified computation compared against measured values that it consumes; never counted as a prediction.
Prospectively fixed conditional test	A diagnostic with a custody-bound target definition, kill band, precision floor, and decision rule fixed before comparison.
Frozen prospective branch prediction	A source calculation and decision rule fixed before any eligible comparison, conditional on a named physical branch. Failure rejects that branch; its scope extends to the whole framework only if the branch is proved forced and exclusive.

assigned to it and enforces their stated boundaries; analytic arguments in this manuscript retain their displayed proofs, and no part of the stack establishes physical realization.

Table 5: Principal verification artifacts and their scope.

Results	**Principal artifacts**
Executable observer, detectors, interacting process and C/E instruments	The observer receipts with exhaustive path replays, seeded executions, feedback controls, exact and interval certificates, and a single verification entry point that replays every receipt and checks its hashes [69, 68]
Overlapping observers, local clocks, screens and the continuum operator	The model packages for the observer correspondence, the compiled clock history, joint and active clocks, repair identification, federated screens and the coherent screen observer, each with exact or interval receipts, independent oracles and replay; the analytic limit arguments with their constants and order of limits [78, 85]
Public-data comparisons	Hash-pinned public records (Bell trials, optical ratio traces, CMB posterior chains) with exact rational reducers, their stated readout hypotheses and replay checks [90, 91]
Normal forms, stability, refinement, complexity and the transactional diamond	The observable-normal-forms library and component-paper reductions; the local-diamond and Newman arguments of this manuscript; the finite transaction-model verifier
Born weights, Lüders selection, Tsirelson saturation, no-cloning	The event-algebra modules for the interface, conditioning, the bound and the public record algebra, with per-module axiom audits
Native quantum functional, modular histories and entropy laws	The closed-history reconstruction and return-census packages; the conditional-resampling and memory-refresh packages with their Lean instances
Charged spectra, conserved quadratics and the mass comparison	Exact integer and interval line tables with independent moment checks; the compiled composite instrument; the source-pinned multiloop replay with its reference tables
Carrier selector and Maxwell selection	Exact sign-matrix enumeration and stability bounds; exact rational rank eliminations with countercontrols for each hypothesis
Rank-three completion	The port-Gram repair band, covariance and frame-quotient modules with a source-pinned receipt and independent verifier
Gauge Lie type, matrix witness, matter image and exterior selection	The screen modules for the carrier action, commutant and incidence response; the exact-arithmetic port current certificate; the trace-balanced kernel and exterior scan modules
Causal precursor, count limit and proper-time readout	The provenance and geometry modules; the source-record family receipts with exact all-pairs counts through $q = 89$ and the sampled $q = 144$ level, the carrier realization receipts and event logs, and the public archive
Null tomography, small-ball arithmetic and tensor completion	The Einstein-branch tensor module for null directions, design, decoder, injectivity and metric ambiguity; explicit null-balance, Ward, Bianchi and scale premises
Angular-rank theorem, frozen branches and the dispersion band	The primitive-port prediction and carrier-class band modules with mutation tests; the custody-bound prediction register
Exact federation and the closure loop	The level-six configuration, seeds, seam arrays, loads, per-schedule ledgers, quotient hashes, standalone verifier and public archive; the level-ten settlement receipts; the event logs and recovered specifications of the canonical and control sources
Dark sector, vacuum coordinate and closure roots	The dark-sector, deep-profile, per-cut and fixed-capacity modules with the matched seen-data diagnostic; the capacity closure and nonidentifiability modules; the interval-certified grain root

15 Discussion and comparison

Operational reconstructions derive quantum structure from information and composition principles [24, 25, 26]. OPH makes a narrower statement at the quantum stage: it takes the finite algebra-state representation as an input and contributes the completed-record surface on which the event, conditioning and correlation identities operate, together with the consensus theorem that defines completion and a positive functional reconstructed from native repair histories. Quantum Darwinism locates classical objectivity in redundant environmental records [51], and the completed-consensus record plays the same public role with redundancy replaced by protected overlap agreement and an explicit repair dynamics. The cellular-automaton programme shares the finite substrate [52], while OPH locates the quantum identities on the completed record surface. Wigner's-friend extensions constrain agents reasoning about one another's unfinished measurements [53], whose premises concern pre-consensus perspectives that the completed surface does not represent. Relational quantum mechanics treats physical facts as relational [49] and the thermal-time programme reads time from modular structure [50]; OPH attaches public facts to protected overlap records and treats modular flow as state- and event-relative. The finite native reference can have trivial or nontrivial modular flow depending on its conditioning; identifying a modular parameter with physical duration requires a common calibrated clock [69]. The local-diamond proof sits beside asynchronous agreement [45, 46], and Dijkstra's self-stabilization separates recovery from continued legitimate operation [47].

The gravitational branch belongs to the thermodynamic and entanglement-based routes to geometry [60, 61, 64], with Bisognano–Wichmann modular flow and half-sided modular inclusions as established analytic ingredients [57, 58, 59]. The route from a null-projected balance and the Bianchi identity to the full equation, with Λ as an integration constant, is familiar from trace-free and unimodular formulations [62, 63]; what is added is the finite nine-direction design with exact constants and the premise contract (G1)–(G6) stated as one implication. The gauge section uses Doplicher–Roberts reconstruction for the sector group [66, 67] and contributes the forcing of the local Lie type from complete response and rechartings inner on that response, a stronger premise than mere internal availability of their implementers.

The closure hypothesis has precedents of a different kind. Wheeler pictured a participatory universe as a self-excited circuit [100]. Tegmark's mathematical-universe hypothesis identifies physical existence with mathematical structure [101], and Nozick proposed that a fundamental principle might subsume itself [102]. OPH adds two demands: the self-subsumption must be inhabited, carried out through records and agents inside the structure, and it must be testable in part, through an executable specification whose observers can be run and whose instruments make predictions. Lawvere's fixed-point theorem shows why unrestricted self-representation fails [103]; the closure condition is restricted to the observable and interventional structure that observers can recover.

A useful comparison of unification proposals asks how much physical structure follows from each declared input and which observations distinguish the resulting models; OPH offers a concrete case with proofs and executable checks attached to individual implications. Formal verification measures deductive reliability, predictive success requires physical realization and eligible measurements, and neither theorem counts nor numerical coincidences define a probability that an architecture describes nature.

16 Conclusion

OPH proposes one closed inhabited history whose observers recover the rules of their world and realize those rules inside it. Protected overlap records supply its notion of a public fact. The proposal brings questions of existence, observer access and physical law into one specification, with a clear acceptance condition: the same source must reproduce the observed joint laws under intervention. The finite closure theorem gives an exact model of that logic on consensus packets; a universe-level fixed point is a further claim. The operational observer specifies access and records, without a theorem about phenomenal character.

The finite mathematical results connect domains usually treated separately. Repair yields canonical public normal forms, and fair-coin repair obeys the entropy laws with its complete tape retained. On a supplied complex algebra-state surface, completed records obey quantum measurement identities. The twelve-port carrier has an intrinsic rank-three slow response, and a faithful complete current with internally implemented rotations has Lie type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$. A specified matter construction attaches those currents, localized modes and a pixel-dependent mass operator to common finite states, and the interacting source has certified charged spectra. Complete source executions test intervention, memory and interaction on that architecture. These statements have explicit premises and reproducible certificates; they preserve the ancestry of their parameters and operators.

Controlled free-field limits, finite interacting responses and a conditional Einstein implication extend the same program toward physics in $3 + 1$ dimensions. A single particle comparison needs a common prepared source and clock, a persistent observable excitation, its physical symmetry and detector identification, and controlled volume and cutoff errors. Two neutral rest energies with a common relativistic dispersion determine a mass ratio without an absolute time calibration. Neither the complete species catalogue nor Einstein dynamics is a prerequisite for that comparison. Full physical composition additionally requires an interacting chiral continuum with the observed massive and massless sectors, and geometry responding to the stress and entropy of those same states. Finite symmetry and topology alone identify neither the measured pole masses nor that responsive geometry.

The scope of OPH is literal: a complete account would explain every public physical fact from one recoverable process, and would make its claims about existence and experience precise at the level they consume. The finite results give that ambition mathematical content; the realization maps and frozen tests state how the larger claim can succeed or fail. Physical adequacy depends on common-source constructions and independent measurements.

Reproducibility statement

All theorems, certificates, simulation configurations, receipts, and the formal library are public in the project repositories [1, 2]. The paper release manifest records the release identifier (r2062) with PDF hashes and sizes; computational receipt bundles pin the applicable source revision, configuration, seed and output hashes, with clean-checkout rebuild instructions and negative controls; and machine-readable claim and prediction provenance accompanies the artifacts [3]. This manuscript synthesizes and strengthens results whose longer proofs also appear in public component preprints on observable normal forms, finite event algebras, consensus, Einstein reconstruction, gauge structure and the executable observer [4, 5, 6, 9, 11, 68], a relationship disclosed here so that overlap is visible.

Statements and Declarations

Funding. The authors declare that no external grants or dedicated third-party research funding were received for the preparation of this manuscript.

Author contributions. Bernhard Mueller developed the OPH concept and mathematical synthesis, wrote the main manuscript, and coordinated the formal and computational evidence. Mario Poneder developed the executable observer construction, including the native observer, the interacting source process and their certificates, and contributed the finite port-current construction and verifier, gauge and matter certificates, and proof review of the edge-entropy and Einstein-normalization branches. Alexander Osika proposed that the screen pixel ratio and the correctable-record capacity are derivable rather than fitted, supplied the detuning fixed-point equation that determines the pixel ratio, and originated the echosahedral patch-hardware design on which the screen-microphysics branch is built, in addition to the physical-realization and hardware boundary, prototype framing, and manuscript review. Kai Xue contributed consistency checks, simulator review, visualization review, and prototype-engineering review. Ben Cassie contributed implementation-claim and physical-evidence review. Peter Nguyen contributed proof auditing, branch-boundary analysis, compact-gauge and global-quotient analysis, and manuscript consistency review. Jinwook Kim developed the constructions, proofs, and counterexamples for the profile partition and structural characterizations, exact behavior-cut recovery, exact finite-state profile transport, adaptive repair stratification, the cumulative attempt-capacity classification, fixed computation-federation progress, and the fixed-federation execution classification, together with the mechanism-variant comparison kernel and conditional example and the source-derived phase-effect selection. He constructed a provably fair emission-order round-robin scheduler with a linear exact-output attempt bound, established order-sharp quadratic accepted-step bounds for the actual compiler with the exponential fanout control, and proved the historical weak-relation fair-stuttering no-go and the weak-fair delay versus bounded-waste horizon separation. He formalized these results in Lean, connected the repair and execution results to the Tower library, and contributed their manuscript, claim registry, ledger and build records. David Matscheko contributed proof review of consensus repair, observable normal forms, modular and Einstein algebra, edge entropy, hypercharge, and the $\mathbb{Z}_6$ quotient. Daniel Ari Friedman contributed independent red-team audits of the papers, the book, the claim and premise ledgers, the Lean library, continuous integration and the custody records, which led to scoped headline claims, displayed values matched to their certified receipts and repaired falsification rules, and the independently reviewed programme-obstruction audit whose controls bound how the carrier, the response triplets, the feedback class and the kinetic term may be attached. Jonathan Hill contributed Lean formalization and proof auditing of observable normal forms, refinement and repair results, complexity classifications, and artifact coverage. William T. Glynn contributed simulator development, computational implementation, reproducible-build infrastructure, and release-validation work in the main OPH research repository. Maarten Antonie Visser contributed consensus-protocol review, physics feedback on the foundations and emergent-spacetime framing, and the source-derived causal-poset reading of authenticated provenance that the event-precedence layer implements. Kale Arnav Anirudha contributed consensus-paper material, asynchronous-agreement analysis, repair-map definitions, and formal review of the protocol surface. Brieuc de La Fournière contributed finite Lean constructions on the source-derived causal-order and carrier-symmetry surfaces, including the cover-relation provenance compiler and the antipodal port-action identification, and external comparison controls for the charged-lepton branch. All authors reviewed and approved the manuscript. All authors agree to be accountable for all aspects of the work.

Corresponding author. Correspondence should be addressed to Bernhard Mueller.

Email: bernhard@floatingpragma.ai.
Data availability. The finite witnesses, simulation configurations, receipts, scientific claim registries, and release manifests supporting this study are available in the public OPH and simulation repositories [1, 2]. Individual receipts and their bundle documentation provide the source and output provenance for computational claims. The artifact map identifies the level-six exact-federation archive and its level-five to level-ten extension, the closure-loop archive, and the causal-poset archive.
Code availability. The Lean sources, exact-arithmetic verifiers, finite scans, simulation code, tests, and rebuild instructions are available in the same public repositories [1, 2]. The computational claims in this manuscript identify their principal files in Section 14.
Ethics approval and consent to participate. Not applicable. The study involved no human participants, human data, or animals.
Consent for publication. Not applicable.
Competing interests. The authors participate in OPH-related research, software, simulation, or prototype-development programs and may receive professional or reputational benefit from this work. These relationships are disclosed. The authors declare no other competing interests.

Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- Icosahedral harmonics, finite-difference dispersion, and nonminimal Standard-Model Extension coefficient searches [111, 113, 109, 110].
- Root lattices, graph incidence, transverse projectors, free-vector oscillators, icosahedral edge harmonics, finite-difference symbols, Lorentz-violating threshold kinematics, and nonminimal Standard-Model Extension coefficient searches [105, 107, 114, 112, 108, 109, 110].
- Spherical-design moment theory, lattice dispersion expansions, and Lorentz-violation coefficient parametrizations [106, 112, 109, 104].

References

[1] FloatingPragma (2026). Observer Patch Holography: Lean library, exact code receipts, finite certificates, and scientific claim registries. https://github.com/FloatingPragma/observer-patch-holography

[2] Mueller, B. (2026). OPH physics simulator and reproduction source. https://github.com/muellerberndt/oph-physics-sim. Exact-federation archive: https://github.com/FloatingPragma/observer-patch-holography/tree/r2062/evidence/exact_federation_L6_canonical_20260909. Closure-loop archive: https://github.com/FloatingPragma/observer-patch-holography/tree/r2062/evidence/closure_loop. Causal-poset archive: https://github.com/FloatingPragma/observer-patch-holography/tree/r2062/evidence/source_net_causal_poset. Active refinement-readout revision: https://github.com/muellerberndt/oph-physics-sim/tree/036b12b608a9ca4c19d69f5e951b3b83a7a51cd8.

[3] FloatingPragma (2026). Precommitted comparison certificates, custody records, and the frozen prediction register. Immutable source receipt: https://github.com/FloatingPragma/observer-patch-holography/blob/66176656dc1143f9ec50ba1a6e409c403545857f/code/a5_fingerprint/runtime/spin_six_primitive_port_prediction_receipt.json. Append-only custody packet: https://github.com/FloatingPragma/observer-patch-holography/tree/bb73f80d459113ccc82d5a5b5219d1dda78e9a79/evidence/custody/falsification/frozen_targets/fz11_2026-07-31. Source-seam edge receipt: https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/code/a5_fingerprint/runtime/seam_current_edge_prediction_receipt.json. Source-seam custody and decision-rule clarification: https://github.com/FloatingPragma/observer-patch-holography/tree/bb73f80d459113ccc82d5a5b5219d1dda78e9a79/evidence/custody/falsification/frozen_targets/fz12_2026-08-02. Koide custody packet: https://github.com/FloatingPragma/observer-patch-holography/tree/bb73f80d459113ccc82d5a5b5219d1dda78e9a79/evidence/custody/falsification/frozen_targets/fz10_2026-07-28. Live register:

https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/claims/frozen_prediction_register.json.

[4] Mueller, B., Kim, J., Matscheko, D., and Hill, J. (2026). *Observation-Determined Normal Forms: Stability, Obstructions, and Refinement in Constraint and Rewrite Systems*. Public manuscript in the same repository revision as this paper. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/extra/observable_normal_forms.pdf

[5] Mueller, B. (2026). *Verified Projection-Event Calculus in Lean 4: Bundled Arbitrary-Partition Pinching, Lüders Retractions, and CHSH Interoperability*. Public manuscript and formal artifact. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/extra/machine_checked_finite_event_algebras.pdf

[6] Mueller, B., Xue, K., Kim, J., Anirudha, K. A., Matscheko, D., and Hill, J. (2026). *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/paper/reality_as_consensus_protocol.pdf

[7] Mueller, B., Osika, A., Xue, K., Cassie, B., de La Fournière, B., and Friedman, D. A. (2026). *Twelve-Port Icosahedral Carriers and Native Seam Repair: The Carrier Layer of Observer Patch Holography*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/paper/screen_microphysics_and_observer_synchronization.pdf

[8] Mueller, B., Poneder, M., Osika, A., Xue, K., Cassie, B., Nguyen, P., and Friedman, D. A. (2026). *Observers Are All You Need*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/paper/observers_are_all_you_need.pdf

[9] Mueller, B., Poneder, M., Osika, A., Xue, K., Nguyen, P., Visser, M. A., Matscheko, D., and de La Fournière, B. (2026). *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/paper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.pdf

[10] Mueller, B., Osika, A., and Friedman, D. A. (2026). *The Fine-Structure Constant as an OPH Pixel Fixed Point*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/extra/fine_structure_constant_derivation.pdf

[11] Mueller, B., Poneder, M., Osika, A., Xue, K., Nguyen, P., Matscheko, D., and Friedman, D. A. (2026). *Standard Model Gauge Structure under Complete-Response and Observer-Transport Premises*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/paper/deriving_standard_model_gauge_structure_from_observer_overlap_consistency.pdf

[12] Mueller, B., Poneder, M., Osika, A., Xue, K., and Friedman, D. A. (2026). *Particles from Native Overlap Transport: Common Currents, Defect Modes and a Pixel-Dependent Mass Operator*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/deriving_the_particle_zoo_from_observer_consistency.pdf

[13] Mueller, B. and de La Fournière, B. (2026). *The Positive-Chamber Koide Identity for Icosahedral Face Circulants*. Public manuscript and formal artifact. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/extra/koide_identity_from_positive_c3_face_circulants.pdf

[14] Mueller, B. (2026). *The de Sitter Time-Advance Sign from a Finite Screen with Fixed Capacity*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/extra/de_sitter_time_advance_sign_from_fixed_screen_capacity.pdf

[15] Mueller, B. and Matscheko, D. (2026). *Observer Patch Holography and the Dark Sector: Modular Charge, the Anomalous Collar Source, and the Deep Galaxy Law*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/cosmology/oph_dark_matter_paper.pdf

[16] Mueller, B. (2026). *Observer Patch Holography Cosmological Vacuum and Structure Formation: Finite Transfer Conditions, Fluctuation Ensembles, and Proto-Objects*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/cosmology/oph_cosmological_vacuum_and_structure_formation.pdf

[17] Mueller, B. (2026). *Observer-Screen Synchronization as a Candidate Inflation-Free Branch: Record-Density Expansion, Required Premises and a Conditional Tilt*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/main/cosmology/oph_inflation_without_inflaton_observer_screen_synchronization.pdf

[18] Mueller, B. (2026). *A Finite Black-Hole Information Ledger for Observer Patch Holography*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/main/cosmology/oph_black_hole_information_ledger.pdf

[19] Koide, Y. (1983). New view of quark and lepton mass hierarchy. *Physical Review D* **28**, 252–254.

[20] Wissner-Gross, A. D. (2026). *A Parameter-Free Candidate for the Fine-Structure Constant from Entangled Qudits*. Preprint. https://alexwg.org/publications/ChainTheoryAlpha.pdf

[21] Mohr, P. J., Newell, D. B., Taylor, B. N., and Tiesinga, E. (2025). CODATA recommended values of the fundamental physical constants: 2022. *Reviews of Modern Physics* **97**, 025002.

[22] Mueller, B. and Hill, J. (2026). *A Conditional Identification of the Yang–Mills Gap with a Finite Repair Gap: Dobrushin Repair Floors and a $\mathbb{Z}_2$ Test of the Transfer Receipt.* Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/extra/yang_mills_gap_clay_problem.pdf

[23] Einstein, A., Podolsky, B., and Rosen, N. (1935). "Can quantum-mechanical description of physical reality be considered complete?" *Physical Review* **47**, 777–780. https://doi.org/10.1103/PhysRev.47.777

[24] Hardy, L. (2001). "Quantum theory from five reasonable axioms." arXiv:quant-ph/0101012. https://arxiv.org/abs/quant-ph/0101012

[25] Chiribella, G., D'Ariano, G. M., and Perinotti, P. (2011). "Informational derivation of quantum theory." *Physical Review A* **84**, 012311. https://doi.org/10.1103/PhysRevA.84.012311

[26] Masanes, L., and Müller, M. P. (2011). "A derivation of quantum theory from physical requirements." *New Journal of Physics* **13**, 063001. https://doi.org/10.1088/1367-2630/13/6/063001

[27] Gleason, A. M. (1957). "Measures on the closed subspaces of a Hilbert space." *Journal of Mathematics and Mechanics* **6**, 885–893.

[28] Busch, P. (2003). "Quantum states and generalized observables: A simple proof of Gleason's theorem." *Physical Review Letters* **91**, 120403. https://doi.org/10.1103/PhysRevLett.91.120403

[29] Clauser, J. F., Horne, M. A., Shimony, A., and Holt, R. A. (1969). "Proposed experiment to test local hidden-variable theories." *Physical Review Letters* **23**, 880–884. https://doi.org/10.1103/PhysRevLett.23.880

[30] Cirel'son, B. S. (1980). "Quantum generalizations of Bell's inequality." *Letters in Mathematical Physics* **4**, 93–100. https://doi.org/10.1007/BF00417500

[31] Landau, L. J. (1987). "On the violation of Bell's inequality in quantum theory." *Physics Letters A* **120**, 54–56. https://doi.org/10.1016/0375-9601(87)90075-2

[32] Wootters, W. K., and Zurek, W. H. (1982). "A single quantum cannot be cloned." *Nature* **299**, 802–803. https://doi.org/10.1038/299802a0

[33] Dieks, D. (1982). "Communication by EPR devices." *Physics Letters A* **92**, 271–272. https://doi.org/10.1016/0375-9601(82)90084-6

[34] Lüders, G. (1950). "Über die Zustandsänderung durch den Meßprozeß." *Annalen der Physik* **443**, 322–328. https://doi.org/10.1002/andp.19504430510

[35] Bombelli, L., Lee, J., Meyer, D., and Sorkin, R. D. (1987). "Space-time as a causal set." *Physical Review Letters* **59**, 521–524. https://doi.org/10.1103/PhysRevLett.59.521

[36] Surya, S. (2019). "The causal set approach to quantum gravity." *Living Reviews in Relativity* **22**, 5. https://doi.org/10.1007/s41114-019-0023-1

[37] Major, S., Rideout, D., and Surya, S. (2007). "On recovering continuum topology from a causal set." *Journal of Mathematical Physics* **48**, 032501. https://doi.org/10.1063/1.2435599

[38] Myrheim, J. (1978). "Statistical geometry." CERN preprint TH-2538.

[39] Meyer, D. A. (1988). "The Dimension of Causal Sets." PhD thesis, Massachusetts Institute of Technology.

[40] Benincasa, D. M. T., and Dowker, F. (2010). "The scalar curvature of a causal set." *Physical Review Letters* **104**, 181301. https://doi.org/10.1103/PhysRevLett.104.181301

[41] Dowker, F., and Glaser, L. (2013). "Causal set d'Alembertians for various dimensions." *Classical and Quantum Gravity* **30**, 195016. https://doi.org/10.1088/0264-9381/30/19/195016

[42] Hawking, S. W., King, A. R., and McCarthy, P. J. (1976). "A new topology for curved space–time which incorporates the causal, differential, and conformal structures." *Journal of Mathematical Physics* **17**, 174–181. https://doi.org/10.1063/1.522874

[43] Malament, D. B. (1977). "The class of continuous timelike curves determines the topology of spacetime." *Journal of Mathematical Physics* **18**, 1399–1404. https://doi.org/10.1063/1.523436

[44] Lamport, L. (1978). "Time, clocks, and the ordering of events in a distributed system." *Communications of the ACM* **21**(7), 558–565. https://doi.org/10.1145/359545.359563

[45] Lamport, L., Shostak, R., and Pease, M. (1982). "The Byzantine generals problem." *ACM Transactions on Programming Languages and Systems* **4**(3), 382–401. https://doi.org/10.1145/357172.357176

[46] Fischer, M. J., Lynch, N. A., and Paterson, M. S. (1985). "Impossibility of distributed consensus with one faulty process." *Journal of the ACM* **32**(2), 374–382. https://doi.org/10.1145/3149.214121

[47] Dijkstra, E. W. (1974). "Self-stabilizing systems in spite of distributed control." *Communications of the ACM* **17**(11), 643–644. https://doi.org/10.1145/361179.361202

[48] Newman, M. H. A. (1942). "On theories with a combinatorial definition of equivalence." *Annals of Mathematics* **43**(2), 223–243. https://doi.org/10.2307/1968867

[49] Rovelli, C. (1996). "Relational quantum mechanics." *International Journal of Theoretical Physics* **35**, 1637–1678. https://doi.org/10.1007/BF02302261

[50] Connes, A., and Rovelli, C. (1994). "Von Neumann algebra automorphisms and time-thermodynamics relation in generally covariant quantum theories." *Classical and Quantum Gravity* **11**, 2899–2917. https://doi.org/10.1088/0264-9381/11/12/007

[51] Zurek, W. H. (2003). "Decoherence, einselection, and the quantum origins of the classical." *Reviews of Modern Physics* **75**, 715–775. https://doi.org/10.1103/RevModPhys.75.715

[52] 't Hooft, G. (2016). *The Cellular Automaton Interpretation of Quantum Mechanics*. Fundamental Theories of Physics 185. Springer, Cham. https://doi.org/10.1007/978-3-319-41285-6

[53] Frauchiger, D., and Renner, R. (2018). "Quantum theory cannot consistently describe the use of itself." *Nature Communications* **9**, 3711. https://doi.org/10.1038/s41467-018-05739-8

[54] Ratcliffe, J. G. (2019). *Foundations of Hyperbolic Manifolds*, 3rd ed. Graduate Texts in Mathematics 149. Springer, Cham. https://doi.org/10.1007/978-3-030-31597-9

[55] Serre, J.-P. (1977). *Linear Representations of Finite Groups*. Graduate Texts in Mathematics 42. Springer, New York. https://doi.org/10.1007/978-1-4684-9458-7

[56] Cohan, N. V. (1958). "The spherical harmonics with the symmetry of the icosahedral group." *Proceedings of the Cambridge Philosophical Society* **54**(1), 28–38. https://doi.org/10.1017/S0305004100033156

[57] Bisognano, J. J., and Wichmann, E. H. (1975). "On the duality condition for a Hermitian scalar field." *Journal of Mathematical Physics* **16**, 985–1007. https://doi.org/10.1063/1.522605

[58] Bisognano, J. J., and Wichmann, E. H. (1976). "On the duality condition for quantum fields." *Journal of Mathematical Physics* **17**, 303–321. https://doi.org/10.1063/1.522898

[59] Wiesbrock, H.-W. (1993). "Half-sided modular inclusions of von Neumann algebras." *Communications in Mathematical Physics* **157**, 83–92. https://doi.org/10.1007/BF02098019

[60] Jacobson, T. (1995). "Thermodynamics of spacetime: The Einstein equation of state." *Physical Review Letters* **75**, 1260–1263. https://doi.org/10.1103/PhysRevLett.75.1260

[61] Jacobson, T. (2016). "Entanglement equilibrium and the Einstein equation." *Physical Review Letters* **116**, 201101. https://doi.org/10.1103/PhysRevLett.116.201101

[62] van der Bij, J. J., van Dam, H., and Ng, Y. J. (1982). "The exchange of massless spin-two particles." *Physica A* **116**, 307–320. https://doi.org/10.1016/0378-4371(82)90247-3

[63] Ellis, G. F. R., van Elst, H., Murugan, J., and Uzan, J.-P. (2011). "On the trace-free Einstein equations as a viable alternative to general relativity." *Classical and Quantum Gravity* **28**, 225007. https://doi.org/10.1088/0264-9381/28/22/225007

[64] Van Raamsdonk, M. (2010). "Building up spacetime with quantum entanglement." *General Relativity and Gravitation* **42**, 2323–2329. https://doi.org/10.1007/s10714-010-1034-0

[65] Hall, B. C. (2015). *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*, 2nd ed. Graduate Texts in Mathematics 222. Springer, Cham. https://doi.org/10.1007/978-3-319-13467-3

[66] Doplicher, S., and Roberts, J. E. (1989). "A new duality theory for compact groups." *Inventiones Mathematicae* **98**, 157–218. https://doi.org/10.1007/BF01388849

[67] Doplicher, S., and Roberts, J. E. (1990). "Why there is a field algebra with a compact gauge group describing the superselection structure in particle physics." *Communications in Mathematical Physics* **131**, 51–107. https://doi.org/10.1007/BF02097680

[68] Mueller, B., Poneder, M., Kim, J., and Friedman, D. A. (2026). *An Executable Observer for Observer Patch Holography: Specification, Memory and Observable Fields*. Public manuscript. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/paper/executable_observer_construction.pdf

[69] FloatingPragma (2026). The runnable OPH model: observer implementation, tests, receipts and construction guide. https://github.com/FloatingPragma/observer-patch-holography/tree/r2062/model

[70] FloatingPragma (2026). Admissible record laws for native repair. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/record_laws/README.md

[71] FloatingPragma (2026). Native quantum mixing and the full-carrier memory boundary. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/QUANTUM_MIXING.md

[72] FloatingPragma (2026). What complete native records can recover. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer/RECORD_RECOVERY.md

[73] FloatingPragma (2026). The infinite-occupation sharp-readout history of the common interacting source. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/INFINITE_HISTORY.md

[74] FloatingPragma (2026). Complete comparisons with the actual finite pointer. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/READOUT.md

[75] FloatingPragma (2026). The finite executable observer specification. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/COMPOSITION.md

[76] FloatingPragma (2026). Coherent native repairs around one continuing quantum history. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/QUANTUM_NATIVE.md

[77] FloatingPragma (2026). Active encoded memory in the continuing observer. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/ENCODED_MEMORY.md

[78] FloatingPragma (2026). One process, overlapping observer records and an effective transfer. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer_correspondence/README.md

[79] FloatingPragma (2026). Source-compiled apparatus in a complete observer-clock history. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/compiled_clock_history/README.md

[80] FloatingPragma (2026). Joint local clocks and quantum fields with explicit memory. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/joint_clock_process/README.md

[81] FloatingPragma (2026). Fixed observer predictions and complete-count comparison: modular clock recovery and native repair response. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/predictions/README.md

[82] FloatingPragma (2026). Identifying the complete current-resolving repair law. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/repair_identification/README.md

[83] FloatingPragma (2026). A local action for native quantum observations. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/native_action/README.md

[84] FloatingPragma (2026). Coherent observer outputs on matched source screens. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/coherent_screen_observer/README.md

[85] FloatingPragma (2026). A spatial fluctuation field from one native history: the common spatial generator, its locality and spectral dimension, and the boundary-trace test. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/physical_space_time/OCCUPATION_FIELD.md

[86] FloatingPragma (2026). Joint time and field laws on matched source screens. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/observer_time_fields/REFINEMENT.md

[87] FloatingPragma (2026). Tangent first laws and finite entropy changes: native reference entropy and region boundaries. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/code/geometry/ENTROPY_FIRST_LAW_AUDIT.md

[88] FloatingPragma (2026). Exact native causal cones. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/common_source/CAUSALITY.md

[89] FloatingPragma (2026). A finite sequential native-law prediction protocol. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/closed_prediction_protocol/README.md

[90] FloatingPragma (2026). Classical record responses and published Bell measurements. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/predictions/BELL_RECORDS.md

[91] FloatingPragma (2026). OPH predictions and public-data comparisons. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/docs/research/PUBLIC_DATA_PREDICTION_AUDIT.md

[92] Hensen, B. et al. (2015). Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres. *Nature* 526, 682–686. https://doi.org/10.1038/nature15759

[93] King, S. A. et al. (2022). An optical atomic clock based on a highly charged ion. *Nature* 611, 43–47. https://doi.org/10.1038/s41586-022-05245-4

[94] Louis, T. et al. (2025). The Atacama Cosmology Telescope: DR6 power spectra, likelihoods and ΛCDM parameters. arXiv:2503.14452. https://arxiv.org/abs/2503.14452

[95] Mosco, U. (1994). Composite media and asymptotic Dirichlet forms. *J. Funct. Anal.* 123, 368–421.

[96] Fukushima, M., Oshima, Y. and Takeda, M. (2011). *Dirichlet Forms and Symmetric Markov Processes*, 2nd edition. De Gruyter, Berlin.

[97] Coulhon, T. and Sikora, A. (2008). Gaussian heat kernel upper bounds via the Phragmén–Lindelöf theorem. *Proc. London Math. Soc.* 96, 507–544.

[98] FloatingPragma (2026). *Selecting one source, interface and repair law.* Model document. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/model/SOURCE_SELECTION.md

[99] FloatingPragma (2026). *The OPH Basis*: the closure hypothesis, the no-outside rule, the agreement condition, the executable specification, and the theorem-library premises. https://github.com/FloatingPragma/observer-patch-holography/blob/r2062/claims/axiom_registry.yaml

[100] Wheeler, J. A. (1983). "Law without law." In J. A. Wheeler and W. H. Zurek (eds.), *Quantum Theory and Measurement*, 182–213. Princeton University Press.

[101] Tegmark, M. (2008). "The mathematical universe." *Foundations of Physics* **38**, 101–150. https://doi.org/10.1007/s10701-007-9186-9

[102] Nozick, R. (1981). *Philosophical Explanations*. Harvard University Press.

[103] Lawvere, F. W. (1969). "Diagonal arguments and cartesian closed categories." *Lecture Notes in Mathematics* **92**, 134–145. https://doi.org/10.1007/BFb0080769

[104] Colladay, D., and Kostelecký, V. A. (1998). "Lorentz-violating extension of the standard model." *Physical Review D* **58**, 116002. https://doi.org/10.1103/PhysRevD.58.116002

[105] Conway, J. H., and Sloane, N. J. A. (1999). *Sphere Packings, Lattices and Groups*. Springer New York. https://doi.org/10.1007/978-1-4757-6568-7

[106] Delsarte, P., Goethals, J. M., and Seidel, J. J. (1977). "Spherical codes and designs." *Geometriae Dedicata* **6**, 363–388. https://doi.org/10.1007/BF03187604

[107] Godsil, C., and Royle, G. (2001). *Algebraic Graph Theory*. Springer. https://doi.org/10.1007/978-1-4613-0163-9

[108] Jacobson, T., Liberati, S., and Mattingly, D. (2003). "Threshold effects and Planck scale Lorentz violation: Combined constraints from high energy astrophysics." *Physical Review D* **67**, 124011. https://doi.org/10.1103/PhysRevD.67.124011

[109] Kostelecký, V. A., and Mewes, M. (2009). "Electrodynamics with Lorentz-violating operators of arbitrary dimension." *Physical Review D* **80**, 015020. https://doi.org/10.1103/PhysRevD.80.015020

[110] Kostelecký, V. A., and Russell, N. (2011). "Data tables for Lorentz and CPT violation." *Reviews of Modern Physics* **83**, 11–31. https://doi.org/10.1103/RevModPhys.83.11

[111] Meyer, B. (1954). "On the Symmetries of Spherical Harmonics." *Canadian Journal of Mathematics* **6**, 135–157. https://doi.org/10.4153/CJM-1954-016-2

[112] Strikwerda, J. C. (2004). *Finite Difference Schemes and Partial Differential Equations*. SIAM. https://doi.org/10.1137/1.9780898717938

[113] Trefethen, L. N. (1982). "Group Velocity in Finite Difference Schemes." *SIAM Review* **24**, 113–136. https://doi.org/10.1137/1024038

[114] Weinberg, S. (1995). *The Quantum Theory of Fields, Volume 1: Foundations*. Cambridge University Press. https://doi.org/10.1017/CBO9781139644167

AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-6 Astra, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.